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**Fieldbus Planning, Installation,
and Service
Manual**

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About This Document

This document describes how to,

- Plan the Fieldbus network
- Plan intrinsically safe systems
- Install and wire Fieldbus hardware in the HPM

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Revision Notes

The following list provides notes concerning all revisions of this document.

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References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Doc ID
High Performance Process Manager Installation	HP20-6xx
High-Performance Process Manager Service	HP13-610
PM I/O Installation	PM20-620

Contacts

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.

Symbol	Definition
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation which, if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING, Risk of electrical shock: Potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.
	ESD HAZARD: Danger of an electro-static discharge to which equipment may be sensitive. Observe precautions for handling electrostatic sensitive devices
	Protective Earth (PE) terminal: Provided for connection of the protective earth (green or green/yellow) supply system conductor.
	Functional earth terminal: Used for non-safety purposes such as noise immunity improvement. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national local electrical code requirements.

Symbol	Definition
	Earth Ground. Functional earth connection. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.
	Chassis Ground: Identifies a connection to the chassis or frame of the equipment shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.

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1. Planning the Fieldbus Network

1.1 Planning Overview

Scope of effort

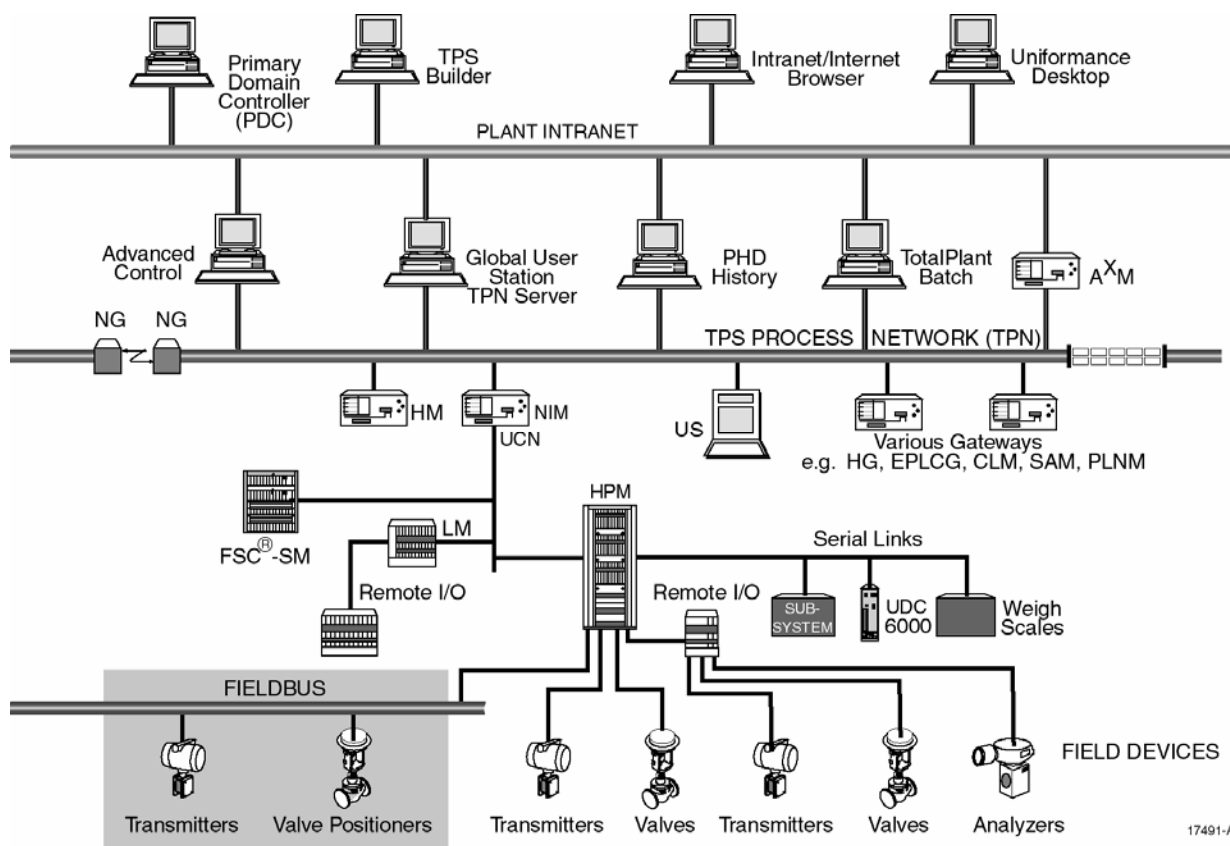
Fieldbus devices connected to TPS Network involve the connection to the following controllers.

- Network Interface Module (NIM)
- High-performance Process Manager (HPM)
- Fieldbus Input Output Processor (FBIOP)
- Fieldbus Field Termination Assembly (FBFTA)
- Intrinsic Safety Field Termination Assembly (ISFTA)

Planning the Fieldbus Network

Planning Overview

Fieldbus in a TPN Network

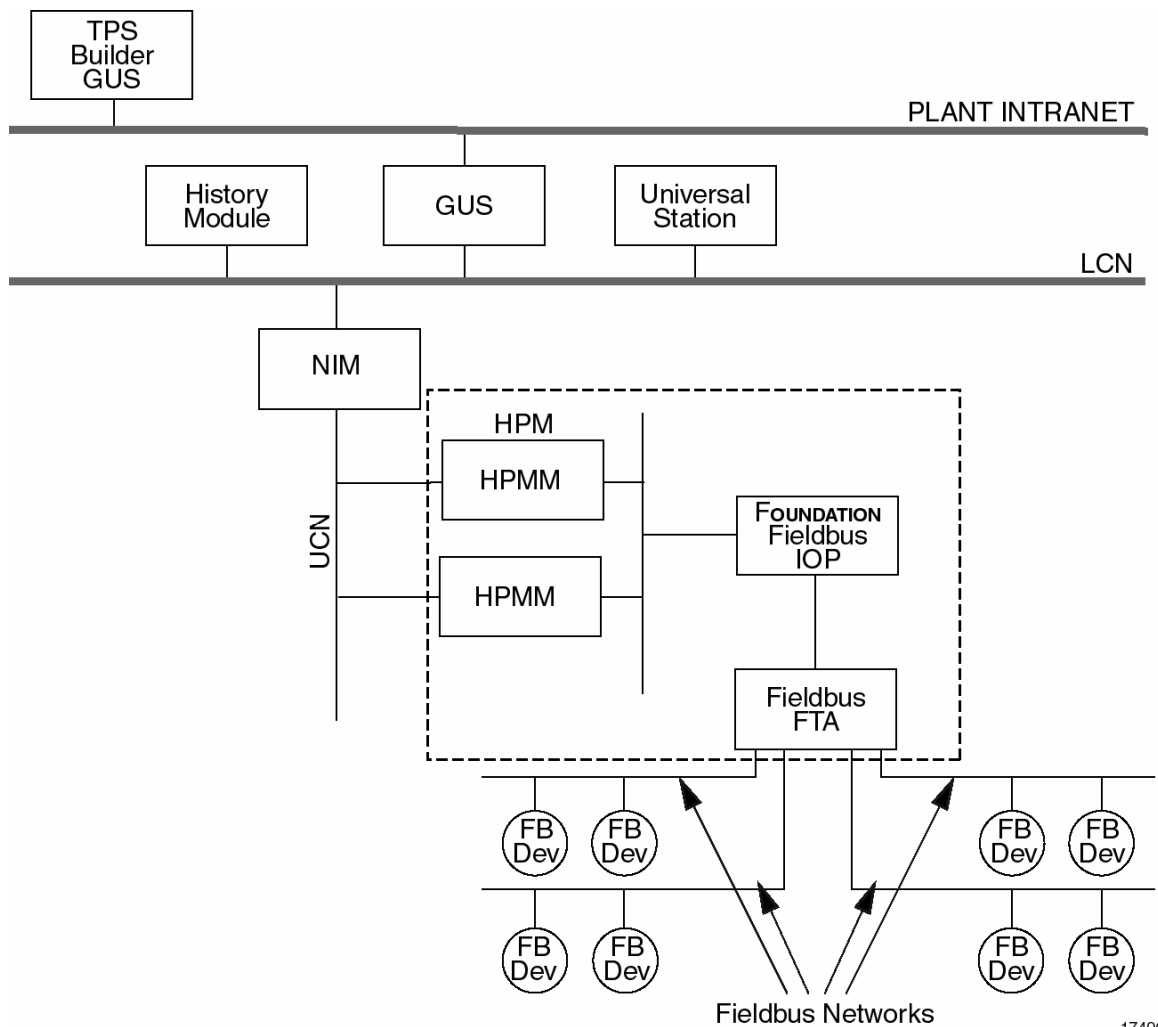


1.2 Components of HPM Fieldbus

Field device illustration

Field devices shown in the following figure are connected to one of the four segments that can be connected to one FBIOP, FBFTA, or ISFTA combination.

Components of HPM Fieldbus



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1.3 Components of a Fieldbus Network

FBIOP purpose

FBIOP interfaces the High Performance Process Manager (HPM) and a single Fieldbus Field Termination Assembly (FTA).

FBIOP FTAs can connect to four fieldbus segments. Each segment can connect to ten devices.

Terminators

An R-C network placed across the fieldbus segment to control the impedance of the circuit within Foundation Fieldbus specifications. For more information see Termination guidelines and Termination placement.

Fieldbus Devices

A sensing device or control device that has a fieldbus interface compatible with Foundation Fieldbus specifications.

Four types of cable

Four types of cable are usable in fieldbus segment wiring, as shown in the following table.

Wire Types and Characteristics

Type	Cable Description	Size	Max Length
A	Shielded, twisted pair	(.8 mm ²) #18 AWG	1900 m (6232 ft.)
B	Multi-twisted-pair with shield	(.32 mm ²) #22 AWG	1200 m (3936 ft.)
C	Multi-twisted-pair without shield	(.13 mm ²) #26 AWG	400 m (1312 ft.)
D	Multi-core, without twisted pairs and having an overall shield	(1.25 mm ²) #26 AWG	400 m (1312 ft.)

New installations

Type A cable is preferred for new installations and is specified in the IEC/ISA Physical Layer Standard, clause 22.7.2.

Existing installation

Type B cable is an acceptable alternate to Type A when multiple fieldbus segment sections are run through the same area of the plant.

Unshielded twisted pair

Type C is a less preferred, single or multiple twisted pair cable used in installations that have been retrofitted for fieldbus. This cable has a limited length that may preclude using it.

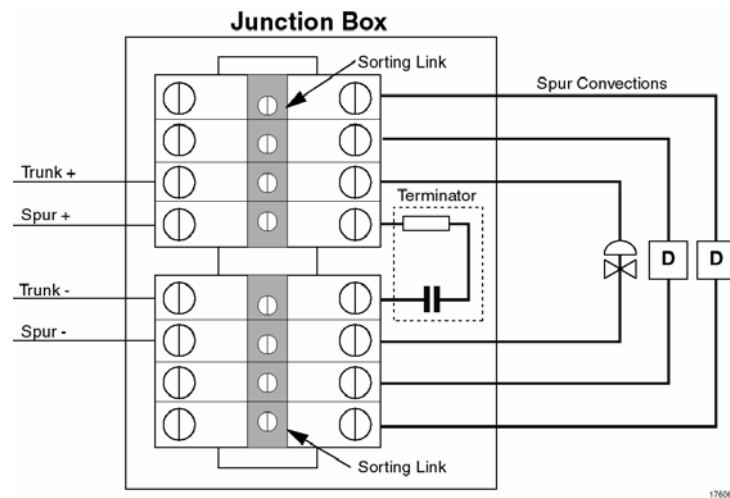
Unshielded multi-core

Type D is the least preferred cable used in installations that have been retrofitted for fieldbus. This cable has a limited length that may preclude using it.

Junction box

An enclosure used for mounting terminal blocks used to connect trunk cables to spur cables.

Junction Box with Terminal Block Wiring



Terminal blocks

Used within a junction box for interconnection of multiple cables, fieldbus devices, and terminators. Today's terminal blocks are designed to facilitate installation and maintenance.

Power supplies

The Fieldbus Foundation Physical Layer Profile Specification has references three types of power supplies.

- | | |
|-----------------|---|
| Type 131 | Non-I.S. power supply intended for feeding an I.S. barrier. Output voltage depends on the barrier rating. |
| Type 132 | Non-I.S. power supply not intended for feeding an I.S. barrier. Output voltage is 32 VDC. |
| Type 133 | I.S. power supply. Complies with the recommended I.S parameters |

The power supply modules used on the FTAs contain the required impedance matching network.

1.4 Fieldbus Segment Configurations

Basic Rules

The following segment configuration rules are mandatory.

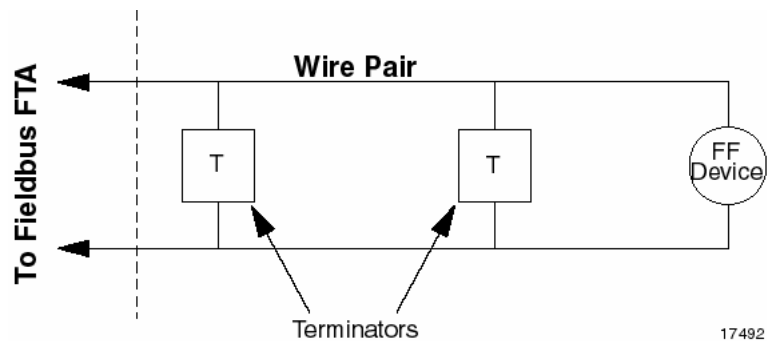
- There must be a terminator at each end of the segment.
- DC power must be provided to each segment either by a power source on the FTA, or by an external power supply.
- Fieldbus devices are limited to ten per segment.

Network configuration types

Network configuration types such as point-to-point, bus with spurs, daisy chain, star (tree), and I.S segments.

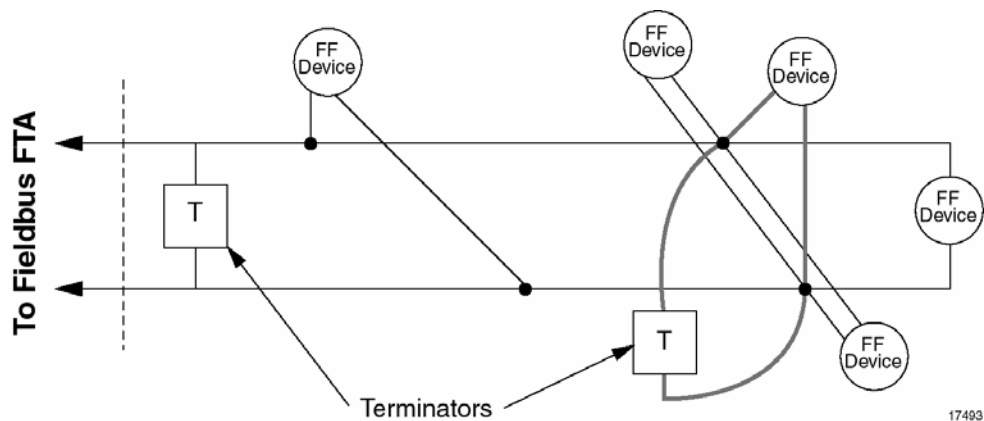
Example of a Basic Fieldbus Loop

This segment has all of the required components. The terminator should be located as close as possible to the end of the trunk cable.



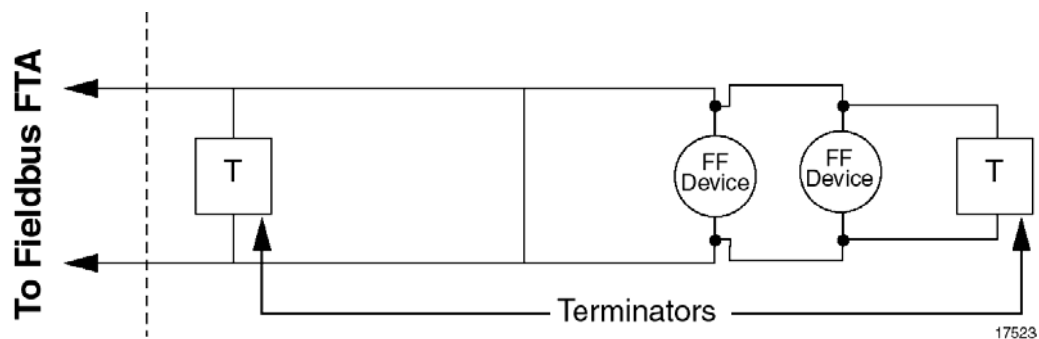
Fieldbus with a Spur and a Star Connection

This is a more complex segment with the terminators as close as possible to the end of the trunk cable.



Fieldbus with a Daisy Chain Connection

With a daisy-chained segment, place the terminator as near to the end of the segment as possible.



1.5 Implementation Guidelines

Number of devices

- The maximum number of devices per segment is 10.
- The maximum number of devices connected to the 4 segments on a single FTA is 40.
- The maximum number of IOPs installed in a single HPM is 10 to 20, depending on the loading of the segments.

Grounding



ATTENTION

The following guidelines are absolute and without exceptions.

- The signal wires are NOT to be connected to ground at any point.
- The cable shields are to be grounded only at the FTA connector.
- Cable shields from multiple cables in the same segment are to be connected together at each terminal block.

1.6 Detailed Planning of Segments

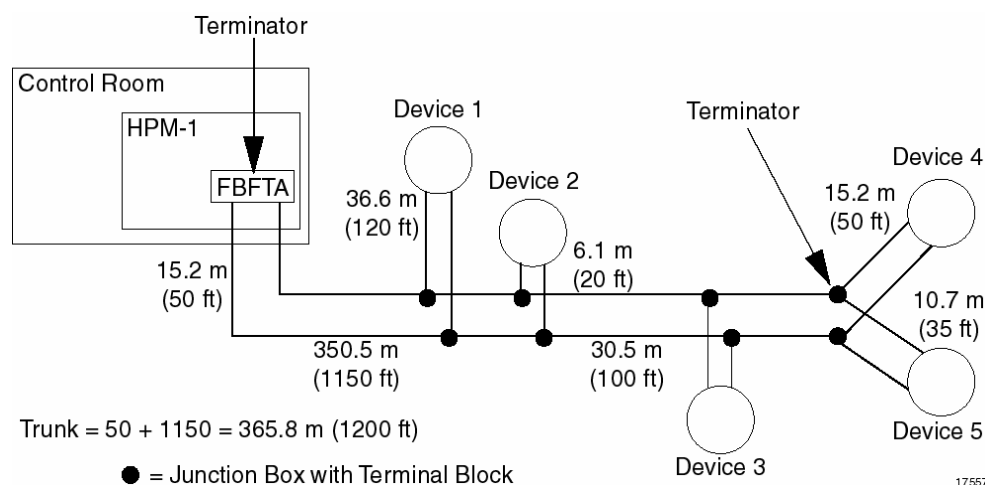
Wiring usage

All new wiring should be done with Type A or Type B wire. Refer to the Wire Types and Characteristics table under the Four types of cable, sub-section for maximum wire lengths (within Components of a Fieldbus Network section.)

Spur segment example

To start optimization of a segment, first plot a trunk cable run that minimizes the trunk cable length versus spur cable length. Calculate the length of the trunk cable. This is the distance (as the wire runs) from the FTA to the farthest end of the segment.

Segment 1 Trunk Length Calculations



If spurs are used, measure the length of cable from the tap terminal to the FF device for each spur. Record lengths for future use.

Segment 1 Cable Length

Cable ID	Cable Type	Length in Meters (Feet)
T001-1 Segment 1 Trunk	Type A	365.8 (1200)
T001-2 Spur 1	Type A	36.6(120)
T001-3 Spur 2	Type A	3.6(20)
T001-3 Spur 3	Type A	30.5(100)
T001-4 Spur-4	Type A	15.2(50)
T001-5 Spur 5	Type A	10.7(35)

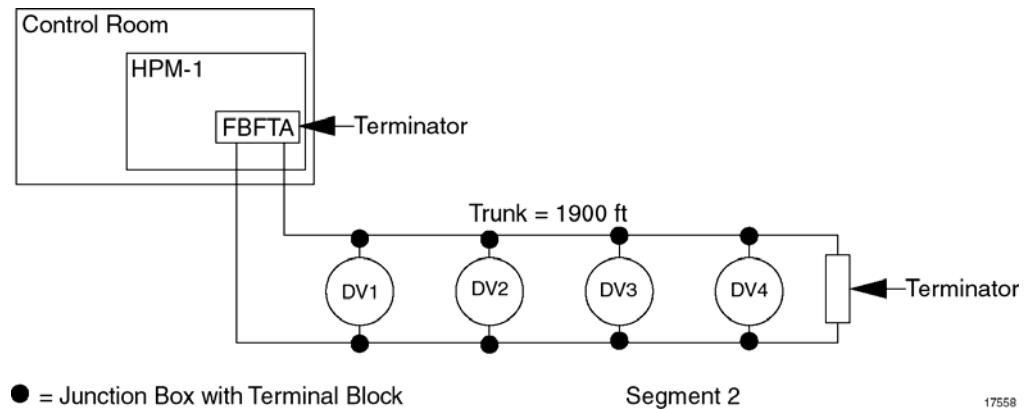
Total = 464.8(1525 ft)*

This example is acceptable if Type A cable is used—Type B is not acceptable. This example could be optimized by rerouting the trunk cable to run more point to point in the vicinity of the devices instead of having a straight-line trunk.

Daisy Chain segment example

To start optimization of a segment, first plot a trunk cable run that minimizes the trunk cable length versus spur cable length. Calculate the length of the trunk cable. This is the distance (as the wire runs) from the FTA to the farthest end of the segment.

Daisy Chain Segment Calculations



Segment 2 Cable Lengths

Cable ID	Cable Type	Length in Meters (Feet)
T001-1 Segment 1 Trunk	Type A	579.1 (1900)

Total length = 579.1 (1900 ft)

This example shows an advantage to daisy-chained segments.

1.7 Planning Segments Using Preexisting Wire

Reuse of wire guidelines

Wire that is already installed can be reused to build a new segment provided the total length of the wire in the segment is within the Four types of cable sub-section (within the Components of a Fieldbus Network section). Wire that does not fall within the descriptions listed in the Wire Types and Characteristics table should not be reused.

Segment planning approach

Plan the segment as if the reused wire is new wire while calculating the formula.

$$LX/MAXX + LY/MAXY < 1$$

Where

Lx = length of cable X

LY = length of cable Y

MAXX = maximum length for X alone

MAXY = maximum length for Y alone

Calculation example

If LX represents 1100 meters (3608 feet) of type A cable mixed with 100 meters (323 feet) of type D cable;

LX = 1100 (3608 feet)

LY = 100 (323 feet)

MAXX = 1900

MAXY = 200

$1100/1900 + 200/200 = 1.079$

The result is >1 so this is not acceptable. A different solution must be found.

Expanded formula

The cable can be expanded for a mixture of the four cable types:

$$LW/MAXW + LX/MAXX + LY/MAXY + LZ/MAXZ <$$

Where

LW = length of cable W

LX = length of cable X

LY = length of cable Y

LZ = length of cable Z

MAXW = maximum length for W alone

MAXX = maximum length for X alone

MAXY = maximum length for Y alone

MAXZ = maximum length for Z alone

1.8 DC Power for Two-Wire Field Devices

Device dependencies

The number of devices on a fieldbus segment is limited depending on the voltage of the power supply, the resistance of the cable and the amount of current drawn by each device.

DC power example

Consider this example:

The power supply and power conditioner output from the FTA is a nominal 19 volts.

- The cable used is 18-gauge wire and has a resistance of 22 ohms/km for each conductor. The home run (trunk) is 1 km long. Therefore, the combined resistance of both wires is 44 ohms.
- Each device draws 20 mA.

Since the minimum voltage needed by each device is 9 volts, there are $19 - 9 = 10$ volts that can be used up by the cable. The total current that can be supplied at the star is

$$\frac{\text{Voltage}}{\text{Resistance}} = \text{Current}$$

$$\frac{10\text{Volts}}{44\text{Ohms}} = 227 \text{ mA}$$

Since each device draws 20 mA, the maximum number of devices at the star of this example is:

$$\frac{227\text{mA}}{20\text{mA}} = 11 \text{ devices}$$

Check device specifications

The power used by fieldbus devices varies by device type and manufacturer. Check the device specifications to determine the device power requirements. One of the gray areas of the power specifications is the initial inrush current and the lift-off voltage. Some

devices may use a great deal more current when they are first turned on and may require more than the 9V minimum voltage to “lift off” and begin operating. The network power distribution calculation should be based on the worst-case inrush current and liftoff voltage numbers. Otherwise the network may not start up when power is first turned on.

1.9 Maximum Trunk and Spur Cable Segments

Trunk segments

The maximum limits indicated in the specification contain a safety factor, to provide assurance that the suggested maximum lengths will not cause communications problems. It is always advisable to keep cable lengths as short as possible.

Spur segments

The maximum allowable spur cable lengths and maximum allowable devices is interactive and dependent upon topology. The following table lists limits of cable length depending upon the number of devices per spur.

Recommended Maximum Spur Length (*m*)

Total Devices	1 Device/Spur	2 Devices/Spur	3 Devices/Spur	4 Devices/Spur
1-9	120 (394)	90 (295)	60 (197)	30 (98)

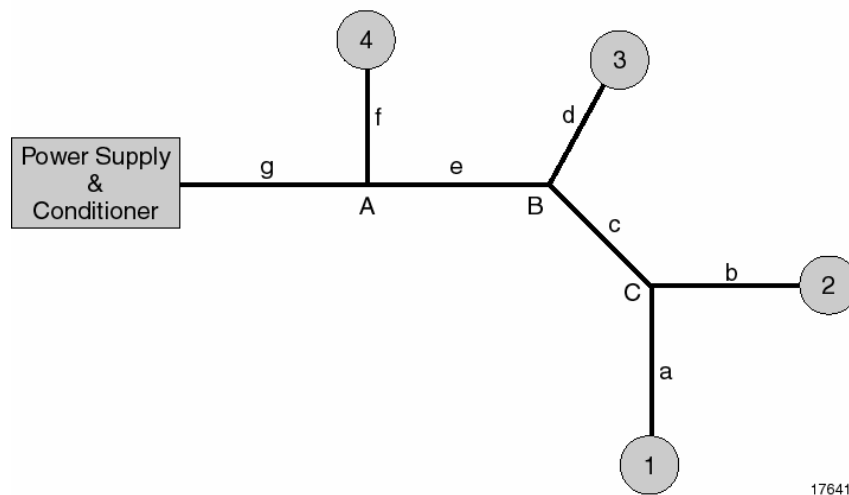
Maximum length of spur cables is the same for all cable types.

1.10 DC Power Calculations with Spur Segments

Rationale for calculation

When a number of devices are on the cable at different places, the power distribution calculation becomes more involved. Following is an example:

Fieldbus Segment with Spurs



Calculate wire resistance

A network is shown with four devices designated 1 through 4. The network wiring has segments a through g. The junctions of the segments are at A, B and C. Here are the facts about the network:

Device	Current Required, mA	Segment	Resistance, Ω
1	20	a	5
2	25	b	10
3	30	c	7
4	15	d	9
		e	6
		f	11
		g	20

Calculate in each segment

From this, the amount of current can be calculated. Starting at the devices furthest from the power source:

Segment	Resistance, Ohms	Current in Segment, mA
a	5	20 (due to device 1)
b	10	22 (due to device 2)
c	7	45 (due to devices +2)
d	9	30 (due to device 3)
e	6	75 (due to devices 1+2+3)
f	11	15 (due to device 4)
g	20	90 (due to device 1+2+3+4s)

Calculate voltage drop in each segment

Because voltage equals resistance times current, the voltage drop in segment can be calculated.

Segment	Resistance, Ohms	Current in Segment, mA	Voltage Drop in Segment ,V
a	5	20 (due to device 1)	0.1
b	10	22 (due to device 2)	0.25
c	7	45 (due to devices +2)	0.315
d	9	30 (due to device 3)	0.27
e	6	75 (due to devices 1+2+3)	0.45
f	11	15 (due to device 4)	0.165
g	20	90 (due to device 1+2+3+4s)	1.8

Voltage drop for each node

From this, the voltage drop at each node can be calculated:

Node	Voltage Drop, V
A	1.8 (due to segment g)
Device 4	1.965 (due to segment g + f)
B	2.25 (due to segment g + e)
Device 3	2.52 (due to segment g e d)
C	2.565 (due to segment g + e + c)
Device 2	2.815 (due to segment g + e + c + b)
Device 1	2.665 (due to segment g + e + c + a)

As the previous table indicates, the largest voltage drop is 2.815 volts at device 2. The current flowing in segment g is 90 mA. Therefore, the power supply and conditioner must be able to deliver at least 90 mA. The lowest voltage that can be at the power supply/conditioner is the 9 volt minimum required by the devices, plus the 2.815 volt drop of the cable segments, plus the 1 volt needed for signaling, plus a safety margin of approximately 1 volt for a total of about 14 volts.

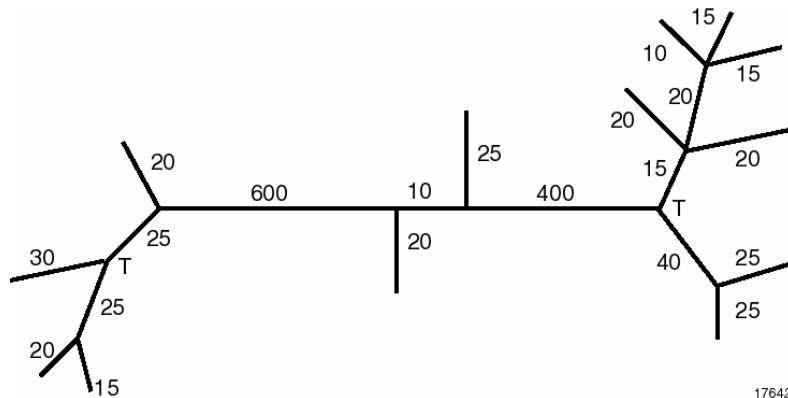
Intrinsic safety barriers can be considered segments in the network with a resistance specified by their manufacturer. This will be covered in more detail later.

1.11 Signal Distortion

Causes

There are many causes for signal distortion. Spurs on the cable are one source. Although it is not possible to provide a definitive analysis of the effects of spurs, here are some guidelines that will help estimate if a particular network will work or is close to having problems. Consider the network below where the lengths of the cable segments are shown in meters.

Example of Segment Spur Length



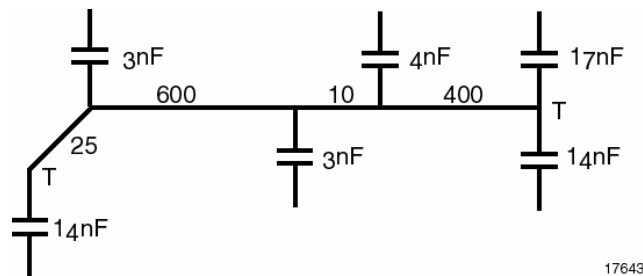
It is not clear in this network which is the home run cable and which are the spurs. In such a network, place the terminators as far away from one another as possible. In this example, they are shown as "T". Consider the cable between the terminators to be the home run cable. Consider all other cables segments as spurs.

Estimate number of spurs

Testing of various cable configurations for signal distortion has shown that spurs up to 300 meters in length do not present a problem. The issue is to determine the allowable number of spurs. A way to estimate this is as follows:

The effect of a spur on the signal is very similar to that of a capacitor (providing the spur is less than 300 meters long). As an estimate, the capacitance of fieldbus cable is about 0.15 nF/meter. In this example, the network can be modeled as the home-run cable with attached capacitors.

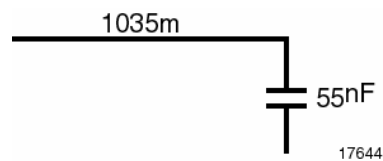
Estimate of Capacitance of Spurs



In this model, the home run cable length is 1035 meters. The capacitance is calculated from the total cable length of each spur.

From the measurements of actual cable, it has been determined that the worst case signal distortion occurs if all the capacitors are on one end of the home run cable. In this example this would be modeled as

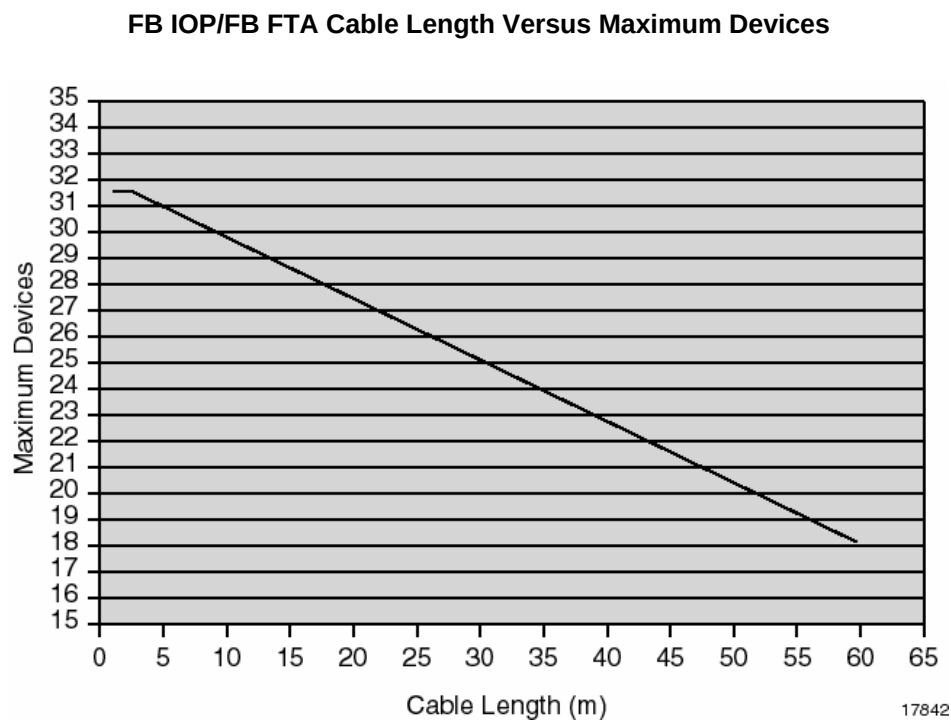
Example of Capacitance with Spurs at the Cable End



1.12 IOP/FTA Cable Length

Maximum devices versus length

The length of the cable that connects to the FBIOP to the FBFTA has an effect on the maximum number of devices that can be installed on a fieldbus segment. As the cable length is increased, the maximum number of fieldbus devices on one segment is decreased as shown in the following figure.



2. Planning Intrinsically Safe Systems

2.1 Intrinsic Safety Guidelines

Use ISFTA in hazardous area



ATTENTION

The following guidelines are absolute and there can not be any exceptions

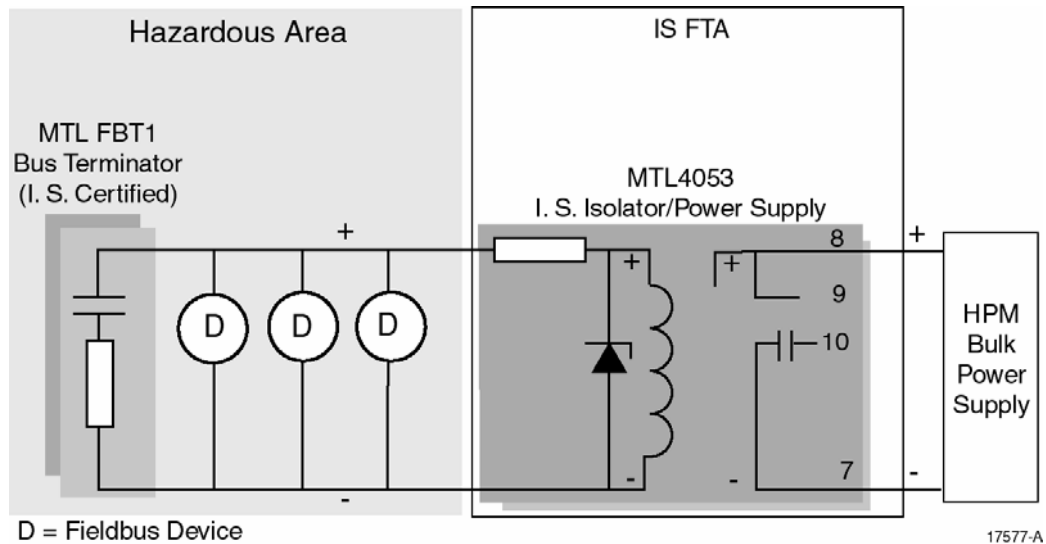
To install a fieldbus segment in a hazardous area, the ISFTA is used in place of the FBFTA discussed in Section 1. The FBIOP, termination, wire length, and wire type are the same. The ISFTA has special circuitry that provides power isolation and limits the amount of electrical current that can be present in the circuit. This provides safety in a hazardous environment.

ISFTA connectivity

The model MU/MC-GFBS01 FTA provides the Fieldbus Galvanic Isolation interface to for fieldbus segments located in an Intrinsic Safety area.

The Intrinsic Safety Field Termination Assembly connects to four hazardous area segments. Each segment can connect to ten devices.

Example of I.S. Segment Wiring



I.S. Wiring guidelines

The wiring guidelines for I.S. segments are the same as for nonincendive segment plus an additional requirement. The available current and voltage drop (dependent on wire length) must be calculated because the MTL4053 circuit limits the power to 80 mA to protect the unit from intermittent or longer-term short circuit faults on the fieldbus segment. See Calculating current available at the end of the cable.

Number of devices

- The maximum number of devices per segment is 10.
- The maximum number of devices connected to the 4 segments on a single FTA is 40.
- The maximum number of IOPs installed in a single HPM is 10 to 20, depending upon how fully loaded the segments are.

Grounding

- The signal wires are NOT to be connected to ground at any point.
- The cable shields are to be grounded only at the FTA connector.
- Cable shields from multiple cables in the same segment are to be connected together at each terminal block.
- Cable shield wires should be insulated within all devices to protect from accidental shorting to ground.

3. Installing the Fieldbus Hardware in the HPM

3.1 Fieldbus Input Output Processor (IOP) Limits

There can be between 10 and 20 FBIOPs installed in one HPM. The limit depends upon the number of fieldbus devices connected to each of the four segments connected to each FTA. Refer to "Subassembly Interconnections" in the High-Performance Process Manager Installation manual for instructions on how to install an IOP.

3.2 Configuration of the Field Termination Assembly (FTA)

Pinning termination jumpers

The MTL4995 provides an internal fieldbus terminator that is actuated by installing the pinning jumper located beside each power supply module on the FTA. If the FTA does not have power supply modules, this pinning will not work—external terminators must be installed.

Termination guidelines

Careful consideration is needed to decide whether it is appropriate to use the terminator within the MTL4995 power supply or to use external bus terminators. The decision depends on the specific wiring and topology of the fieldbus segment. By choosing where to place terminators on the fieldbus segment, the user defines the position and length of the main bus trunk cable and any spur cables present. Clause 22 of the IEC 1158-2 standard specifies some simple rules that guarantee correct operation of the Physical Layer.

In summary, these are:

- The maximum cable length between any two devices within one fieldbus segment should not exceed 1900 m. (If a longer distance than this is required the repeaters may be used.)
- Two bus terminators (each consisting of a 100 Ω resistor and a 1 μ F capacitor in series) should be placed on each fieldbus segment.
- The section of the fieldbus cables between the two bus terminators is called the main bus trunk cable.
- Cable runs to devices not connected within the main bus trunk cable are called spur cables. These have recommended maximum lengths depending on the number of devices connected to them.

The maximum number of devices that can be connected to a segment is 10.

Termination placement

The connection point of the MTL4995 power supply with its internal terminator connected can therefore define the positions of these parts of the fieldbus segment. The figures in Field device illustration, Example of a Basic Fieldbus Loop and Fieldbus with a Spur and a Star Connection contain simple examples of a segment to illustrate the basic principles involved.

Spur cables connecting to a single fieldbus device can generally be a maximum of 120m long, but should be shorter than this if they connect to multiple devices. If this limit, or the overall cable distance between devices are exceeded, then an unacceptable level of communication errors may occur. Hence the correct placement of bus terminators is an important factor to consider.

3.3 Installation of the Field Termination Assembly (FTA)

Installation summary

The technology of installing the FBIOP and FBFTA is documented in Section 9 of HPM Installation, HP20-600.

- The FBIOP installable in any I/O slot.
- The FBFTA is a size B FTA and is installed as a standard size B FTA.
- The Intrinsic Safety FBIOP is installed as another I.S. FTA.
- The standard IOP-to-FTA cable is used to connect the IOP to the FTA.

CE Compliance

All the assemblies that constitute the FOUNDATION Fieldbus (FBUS) are CE Compliant.

Nonincendive

Some FBUS FTAs are approved as Nonincendive for use in Division 2 (Zone 2) hazardous areas. These are models MU/MC-TFBS01 and MU/MC-TFBS11.

Intrinsic Safety

The MU/MC-GFBS01 FBUS FTA is approved as Intrinsically Safe for use in Division 1 (Zones 0 and 1) hazardous areas.

Division 2 (Zone 2) mounting

All IOP and FTA models can be mounted in Division 2 (Zone 2) areas.

FTA description

The fieldbus IOP and FTA together provide an interface for a type H1 (31.25 kbit/second) Foundation fieldbus. Up to four fieldbus segments (buses/networks) can be connected to a single FTA.

The model MU/MC-GFBS01 fieldbus FTA provides galvanic isolation between the field wiring and the FTA.

Depending upon current requirements and the FTA model used, up to ten devices can be connected to each fieldbus segment.

Fieldbus FTA mounting

The fieldbus FTA is a B-size (12-inch) FTA and is approved for mounting in Division 2 (Zone 2) areas.

Fieldbus assemblies

The fieldbus assembly models with their part numbers are listed in the following table.

Fieldbus and IOP Models

Model Number	Description	Part Number
MU-TFBS01	Fieldbus FTA with Power Supply Modules (Removable Screw Terminals)	51309254-125
MC-TFBS01	Fieldbus FTA with Power Supply Modules (Removable Screw Terminals)—Conformally Coated	51309254-175
MU-TFBS11	Fieldbus FTA without Power Supply Modules (Removable Screw Terminals)	51309254-325
MC-TFBS11	Fieldbus FTA without Power Supply Modules (Removable Screw Terminals)—Conformally Coated	51309254-375
MU-GFBS01	Fieldbus FTA with GI/IS Power Supply Modules (Compression Terminals)	51309254-225
MC-GFBS01	Fieldbus FTA with GI/IS Power Supply Modules (Compression Terminals)—Conformally Coated	51309254-275
MU-PFBS01	Fieldbus IOP	51309510-125

Installing the Fieldbus Hardware in the HPM
Installation of the Field Termination Assembly (FTA)

MC-PFBS01	Fieldbus IOP—Conformally Coated	51309510-175
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Nonincendive application for Fieldbus FTA with power supply modules

The following figure illustrates a Nonincendive application for a fieldbus FTA with a power supply module for each segment (model MU/MC-TFBS01).

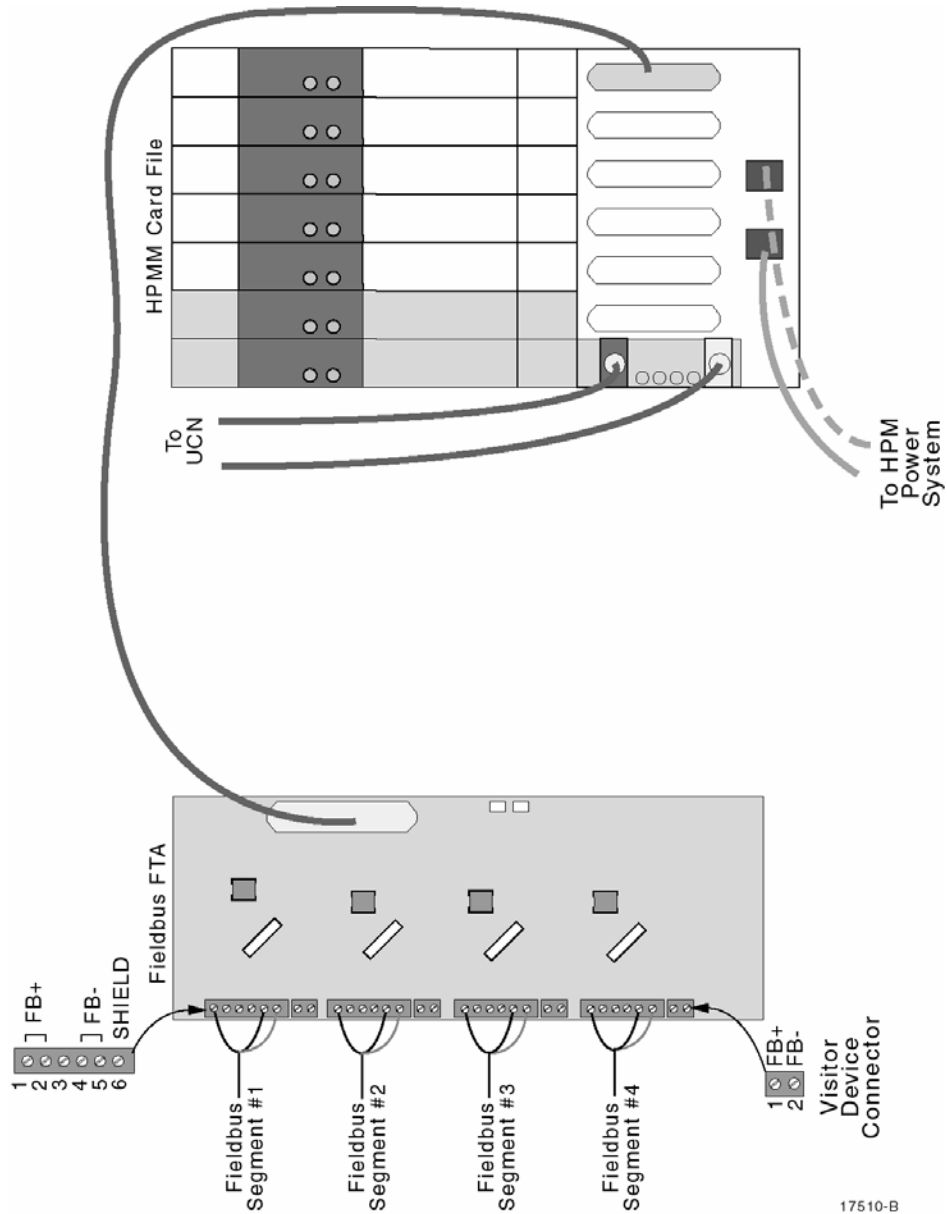
Interconnections for FBFTA with Power Supply Modules Nonincendive Configuration



Nonincendive application for Fieldbus FTA without power supply modules

The following figure illustrates a Nonincendive application for a fieldbus FTA without a power supply module for each segment (model MU/MC-TFBS11). Power for each bus segment must be provided by an external power source.

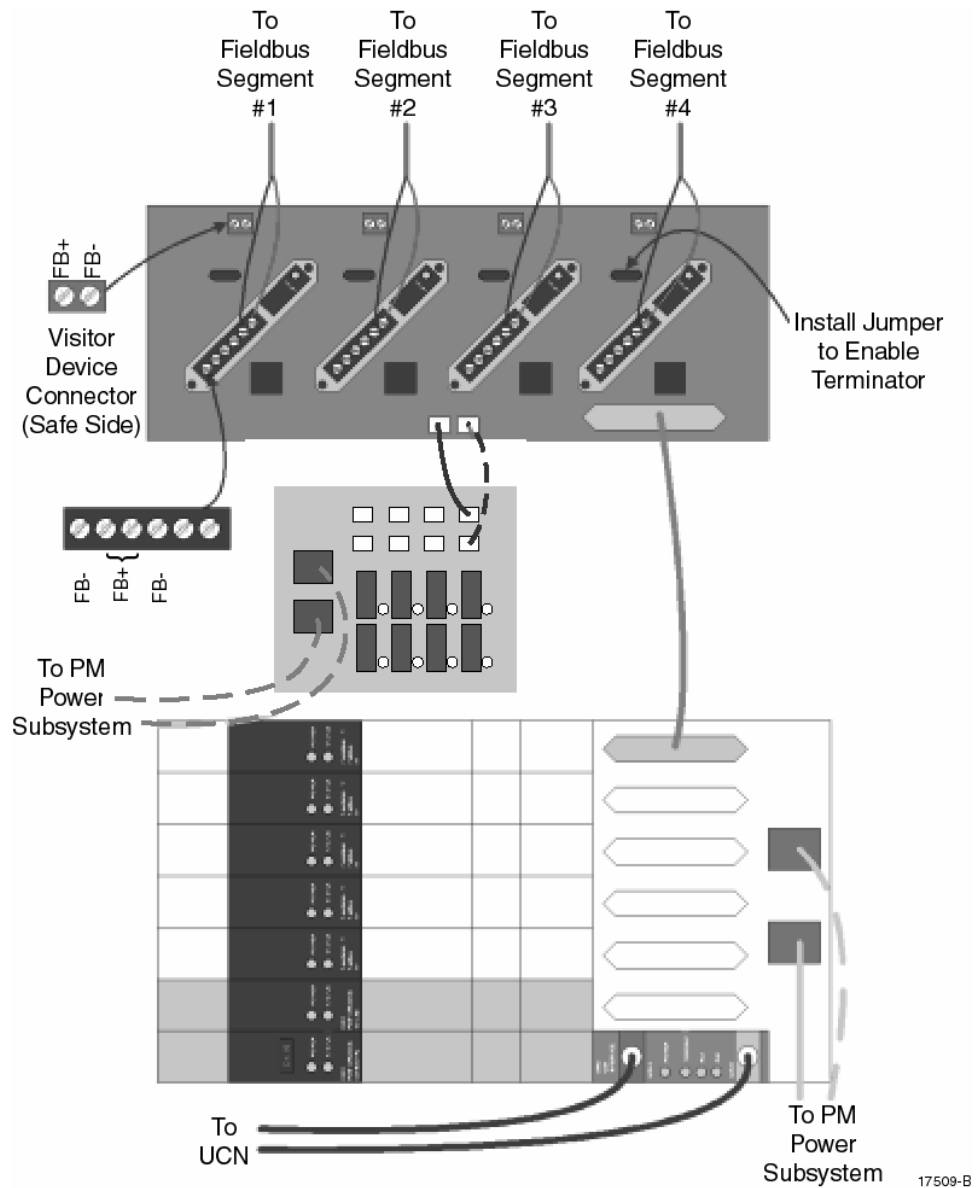
FB FTA without Power Supply Modules Nonincendive Configuration Interconnections



Intrinsically Safe application for Fieldbus FTA with GI/IS power supply modules

The following figure illustrates a Galvanically Isolated/Intrinsically Safe application for a fieldbus FTA with a GI/IS power supply module for each segment (model MU/MC-GFBS01).

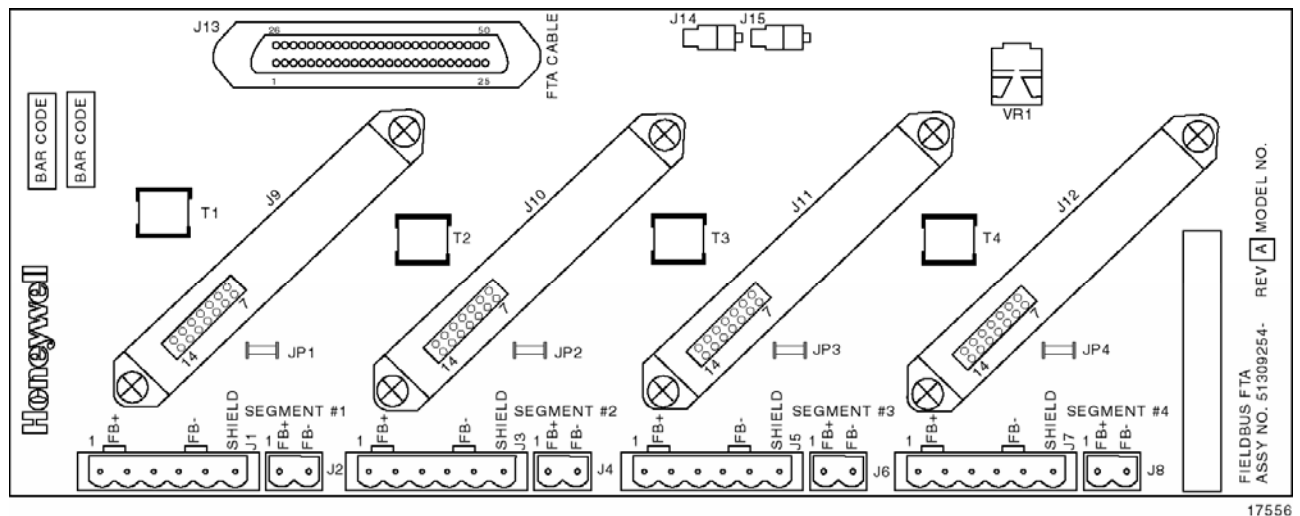
**FB FTA with GI/IS Power Supply Modules I. S. Configuration
Interconnections**



Fieldbus FTA assembly layout

The following figure illustrates the assembly layout of the fieldbus FTA assembly. How J9, J10, J11, and J12 are populated with power supply modules, or not populated with power supply modules, is dependent upon the FTA's model number. Whether the 6-pin connectors are populated is also dependent on the model number. For model descriptions, see the Fieldbus and IOP Models table in the Fieldbus assemblies section.

Fieldbus FTA Assembly Layout



3.4 IOP to Fieldbus FTA Cabling

Purpose of IOP cable

Two areas require attention here:

- These cables are for communication signaling, not power. The power connections are made by the power distribution cable (MU-KGPRXX) to the FTA.
- IOP to FTA cable lengths affect fieldbus unit load characteristics. For FTA cable lengths greater than 5 meters, refer to the FB IOP/FB FTA Cable Length Versus Maximum Devices figure in Maximum devices versus length. This will affect the total number of physical devices that can be connected to a single fieldbus segment. The specification allows 32 physical device unit loads per segment.

Non-CE Compliant IOP to FTA cabling application

For a non-CE Compliant application, the IOP to FTA power connection can be made by using a model MU-KFTAxx cable (the suffix “xx” represents the length of the cable in meters) in lengths up to 50 meters (164 feet) in length. These cables are illustrated in the figures in Nonincendive application for Fieldbus FTA without power supply modules, Nonincendive application for Fieldbus FTA with power supply modules, and Intrinsically Safe application for Fieldbus FTA with GI/IS power supply modules.



REFERENCE - EXTERNAL

See the *High-Performance Process Manager Service* or the *High-Performance Process Manager Installation* manuals for the cable lengths that are available.

CE Compliant IOP to FTA cabling application

For a CE Compliant application, the IOP to FTA power connection can be made by using a model MU-KFTSxx cable (the suffix “xx” represents the length of the cable in meters) in 12 lengths, up to 50 meters (164 feet) in length. These cables are illustrated in the figures in Nonincendive application for Fieldbus FTA without power supply modules, Nonincendive application for Fieldbus FTA with power supply modules, and Intrinsically Safe application for Fieldbus FTA with GI/IS power supply modules.

CAUTION Please be aware that the IOP to FTA cable is not shielded.



REFERENCE - EXTERNAL

See the *High-Performance Process Manager Service* or the *High-Performance Process Manager Installation* manuals for the cable lengths that are available.

3.5 GI Power Distribution to Fieldbus FTA Cabling

Assembly description

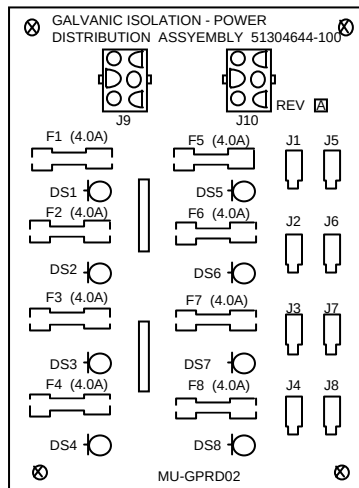
The model MU/MC-GPRD02 GI Power Distribution Assembly (following figure) is required to provide individually fused 24 Vdc power for a fieldbus FTA. The dc power

Installing the Fieldbus Hardware in the HPM

GI Power Distribution to Fieldbus FTA Cabling

is supplied to the GI Power Distribution Assembly by the same type of 6-pin connector cable, model MU-KDPRxx or MU-KSPRxx (the suffix “xx” represents the length of the cable in meters), used to supply power to the card files from the HPM Power System. The GI Power Distribution Assembly has two input power connectors that provide for redundancy cables and protects the system from a single cable or connector failure.

Model MU/MC-GPRD02 Galvanic Isolation Power Distribution Assembly Layout



FTA power cable connections

Redundant model MU-KGPRxx power cables connect to J14 and J15 on the fieldbus FTA and provide up to 4 amperes of 24 V dc fuse-protected power to the FTA. For the location of J14 and J15. Refer to the figures in Nonincendive application for Fieldbus FTA with power supply modules and Nonincendive application for Fieldbus FTA without power supply modules. Only one cable is required.

4. Installation of Field Wiring

4.1 About This Section

Scope

The connection illustrations in this section, together with the illustrations in "Planning the Fieldbus Network" and "Planning Intrinsically Safe Systems" enable the correct wiring of the Foundation Fieldbus segment incorporating all the required elements for the requisite operation.

4.2 Model MU/MC-TFBS01 Fieldbus FTA

Description of FTA

Up to 4 fieldbus segments (buses/networks) can be connected to the model MU/MC-TFBS01 FTA;. The segments are identified as 1 through 4 on the assembly. Each segment has dedicated 6-pin and 2-pin removable screw connectors and a bus terminator enable jumper socket. The bus terminator is located inside the power supply module. The 2-pin connector provides a convenient way of communicating with the bus segment through a visitor device (see **Attention** below).

Controlled power is provided to the segment buses by the power supply modules (MTL4995) that are installed in socket J9 for Segment 1, socket J10 for Segment 2, socket J11 for Segment 3, and socket J12 for Segment 4.

Field wiring connector descriptions

The following table lists the field wiring connections for Segment 1's 6-pin and 2-pin connectors, labeled J1 and J2, respectively. Connections for Segments 2, 3, and 4 are identical, except the connectors are labeled J.Cold Junction Reference;3 and J4, J5 and J6, and J7 and J8, respectively.

Fieldbus FTA Field Wiring Connectors

Connector/Terminal	Terminal Label	Description
J1-1	FB+	Positive fieldbus connection
J1-2	FB+	Positive fieldbus connection
J1-3		Not Used
J1-4	FB-	Negative fieldbus connection

Installation of Field Wiring

Model MU/MC-TFBS01 Fieldbus FTA

Connector/Terminal	Terminal Label	Description
J1-5	FB-	Negative fieldbus connection
J1-6	Shield	fieldbus cable shield connection
J2-1	FB+	Positive Visitor device. Communicator connection
J2-2	FB-	Negative Visitor device Communicator connection



ATTENTION

A visitor device is a device used to analyze, test, or troubleshoot fieldbus devices on a single segment. It can be an analyzer, an external computer or any device that is not part of the process control network, which can read or analyze fieldbus data.

Fieldbus segment terminator jumper

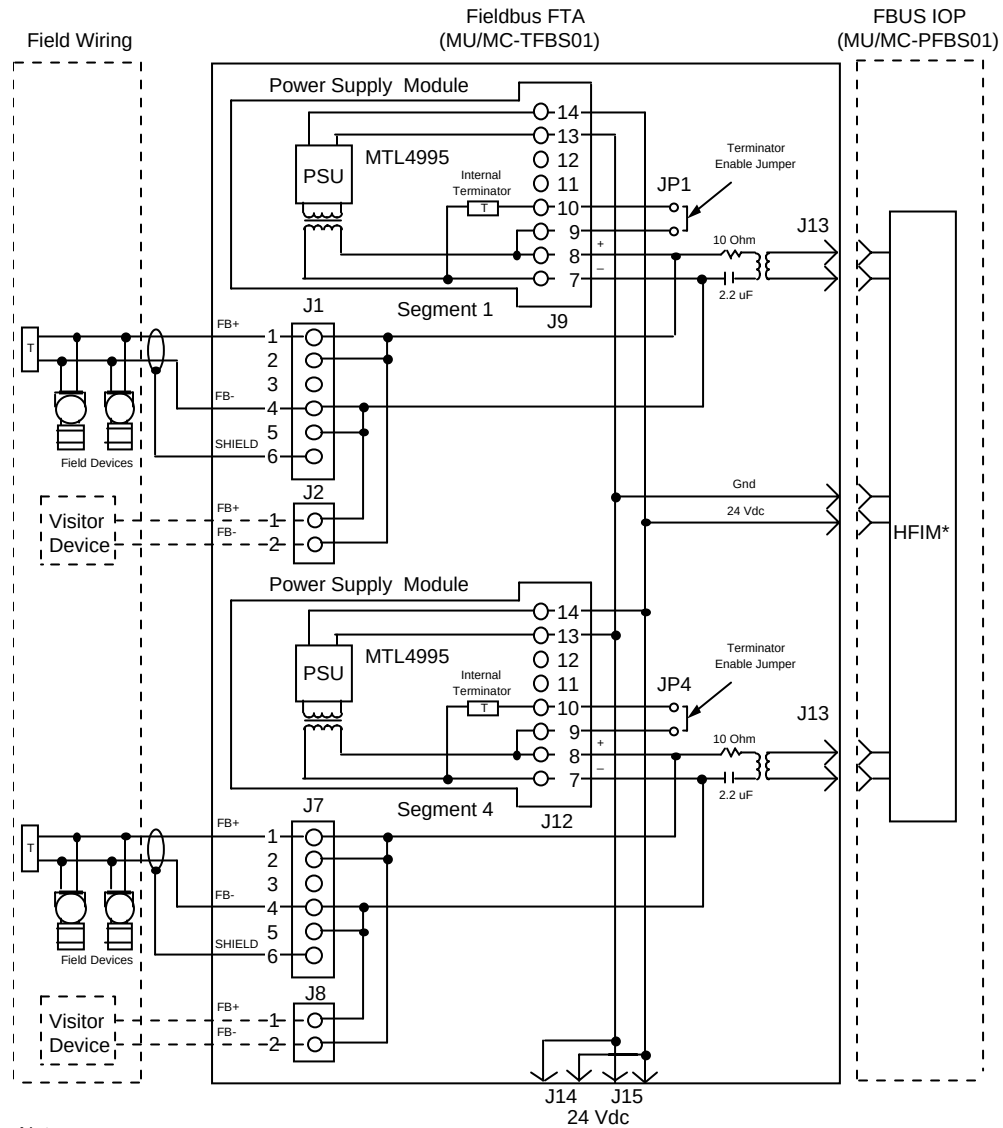
The socket for Segment 1's bus terminator enable jumper socket is labeled JP1. The sockets for Segments 2, 3, and 4 are labeled JP2, JP3, and JP4, respectively. Four jumpers (30731551-001) are included with the FTA assembly.

Model MU/MC-TFBS01 connection diagram

The following figure is a connection diagram for the model MU/MC-TFBS01 fieldbus FTA.

Model MU/MC-TFBS01 Fieldbus FTA Connection Diagram

Installation of Field Wiring Model MU/MC-TFBS01 Fieldbus FTA



Notes:

From GI Power Distribution Assembly

* HFIM = Honeywell Fieldbus Integrated MAU (Medium Attached Unit)
The Power Supply Module is Honeywell part number 51192538-100 (MTL4995).
Connections for Segments 1 and 4 are shown. Segments 2 and 3 are identical.

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4.3 Model MU/MC-TFBS11 Fieldbus FTA Field Wiring

Description of FTA

Up to 4 fieldbus segments (buses/networks) can be connected to the model MU/MC-TFBS11 FTA;. The segments are identified as 1 through 4 on the assembly. Each segment has dedicated 6-pin and 2-pin removable screw connectors. The 2-pin connector provides a convenient way of communicating with the bus segment through a visitor device communicator.

Field wiring connector descriptions

The Fieldbus FTA Field Wiring Connectors table in Model MU/MC-TFBS01 Fieldbus FTA lists the field wiring connections for Segment 1's 6-pin and 2-pin connectors, labeled J1 and J2, respectively. Connections for Segments 2, 3, and 4 are identical, except the connectors are labeled J.Cold Junction Reference;3 and J4, J5 and J6, and J7 and J8, respectively.

Fieldbus segment terminator jumper

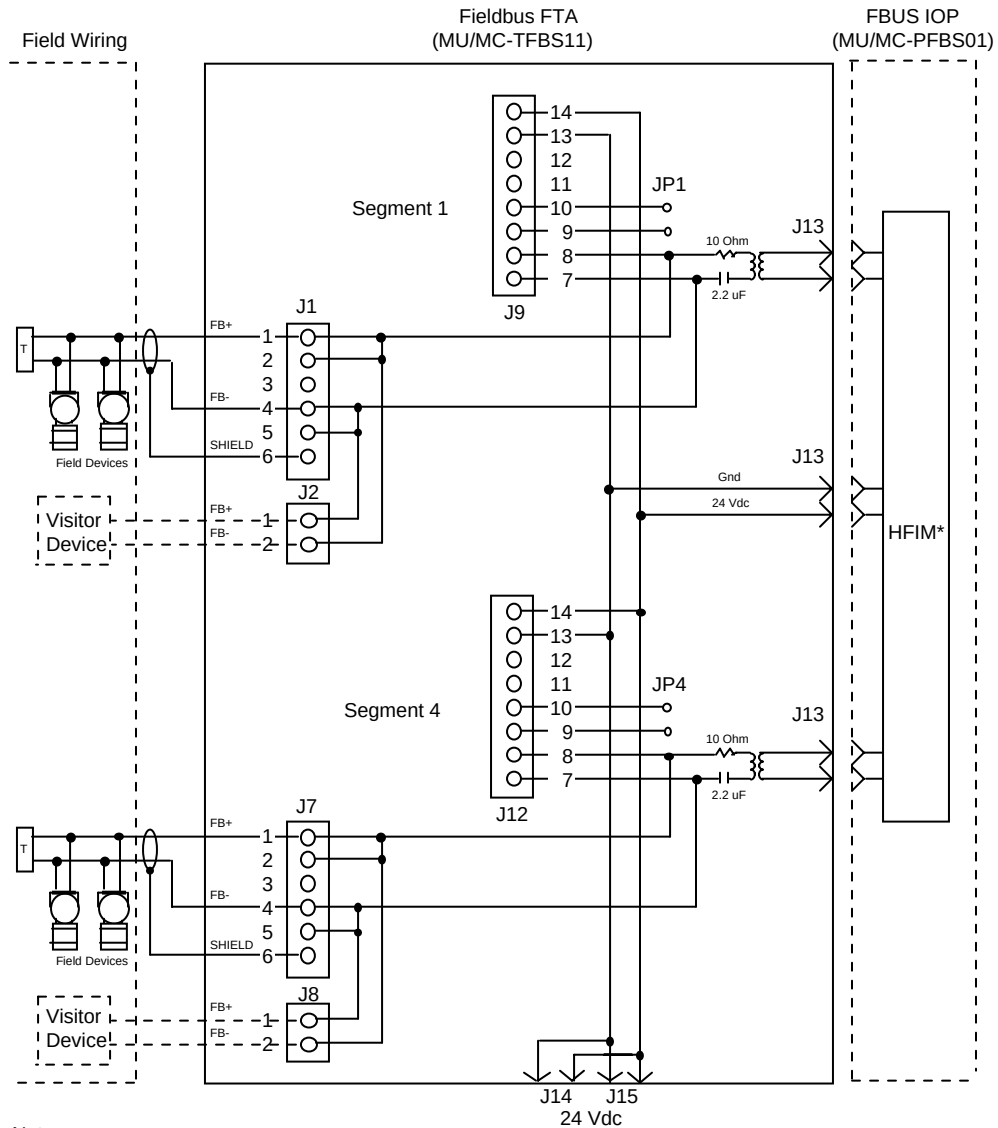
Unlike the model MU/MC-TFBS01 FTA, the segment buses cannot be terminated on the FTA assembly by installing a jumper (30731551-001) in sockets JP1, JP2, JP3, and JP4 because J9, J10, J11, and J12 are not populated with power supply modules.

Model MU/MC-TFBS11 connection diagram

The following figure is a connection diagram for the model MU/MC-TFBS01 fieldbus FTA.

Model MU/MC-TFBS11 Fieldbus FTA Connection Diagram

Installation of Field Wiring Model MU/MC-TFBS11 Fieldbus FTA Field Wiring



Notes:

From GI Power Distribution Assembly

* HFIM = Honeywell Fieldbus Integrated MAU (Medium Attached Unit)
The Power Supply Module is Honeywell part number 51192538-100 (MTL4995).
Connections for Segments 1 and 4 are shown. Segments 2 and 3 are identical.

17600-A

4.4 Model MU/MC-GFBS01 Fieldbus FTA Field Wiring

Description of FTA

Up to 4 fieldbus segments (buses/networks) can be connected to the model MU/MC-GFBS01 FTAs. The segments are identified as 1 through 4 on the assembly. Each segment has dedicated 6-pin compression terminal connector and 2-pin removable screw connectors and a bus terminator enable jumper socket. The bus terminator is located inside the power supply module. The 2-pin connector provides a convenient way of communicating with the IOP side of the GI/IS power supply module (MTL4053) through a visitor device communicator.

Power is provided to the segment buses by the GI/IS power supply modules that are located in socket J9 for Segment 1, socket J10 for Segment 2, socket J11 for Segment 3, and socket J12 for Segment 4.

The model MU/MC-GFBS01 FTA connector has compression terminals.



REFERENCE - EXTERNAL

For a description of the connectors and their use see "Galvanically Isolated Installation", "Field Wiring Connections", in the *PM I/O Installation* manual.

Field wiring connector descriptions

The following table lists the field wiring connections for Segment 1's 6-pin and 2-pin connectors. The 6-pin connector is located on the GI/IS power supply module at location J9. The 2-pin connector is labeled J2.

Connections for Segments 2, 3, and 4 are identical, except the 6-pin connectors are located on the GI/IS power supply module at location J10, J11, and J12, respectively. The 2-pin connectors are labeled J4, J6 and J8, respectively.

MU/MC-GFBS01 FB FTA Field Wiring Connectors

Module/ Terminal	Terminal Label	Description
J9-1		Negative (FB-) fieldbus connection
J9-2		Positive (FB+) fieldbus connection
J9-3		Positive (FB+) fieldbus connection
J9-4		Negative (FB-) fieldbus connection
J9-5		Not Used
J9-6		Not Used
J2-1	FB+	Positive Visitor Device Communicator connection
J2-2	FB-	Negative Visitor Device Communicator connection

Fieldbus segment terminator jumper

The socket for Segment 1's bus terminator enable jumper socket is labeled JP1. The sockets for Segments 2, 3, and 4 are labeled JP2, JP3, and JP4, respectively. Four jumpers (30731551-001) are included with the FTA assembly.

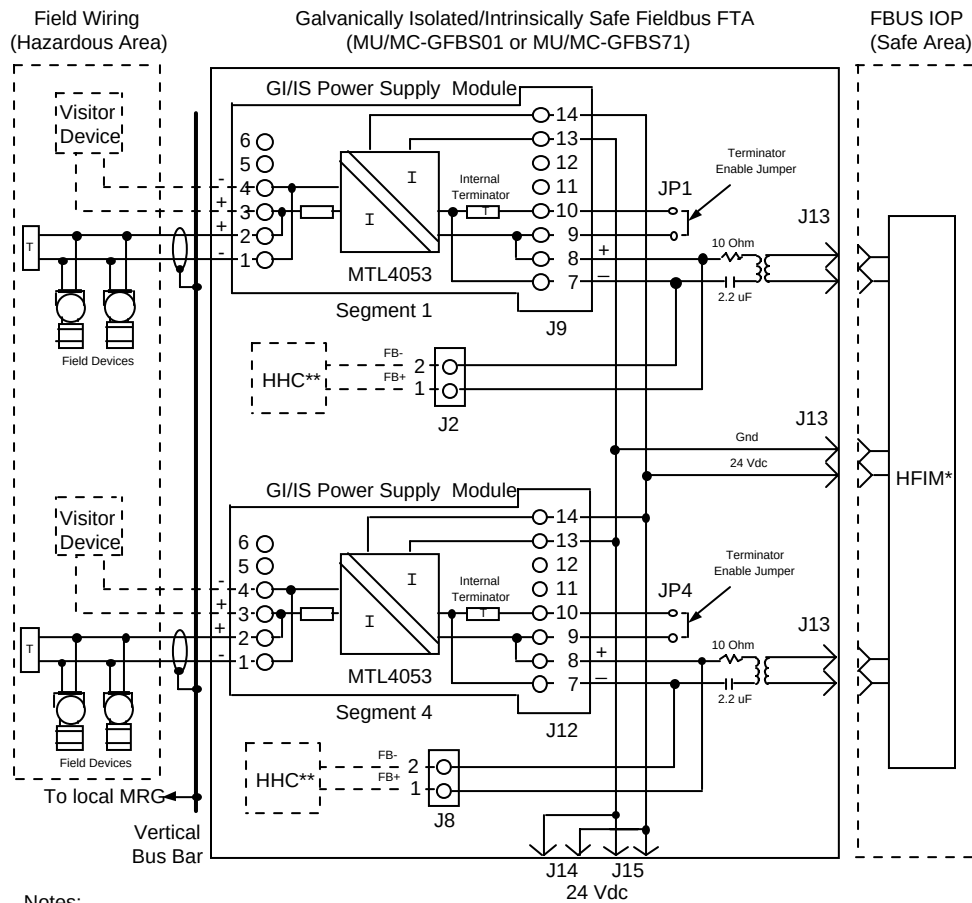
Model MU/MC-GFBS01 FTA connection diagram

The following figure is a connection diagram for the model MU/MC-GFBS01 fieldbus FTAs.

Installation of Field Wiring

Model MU/MC-GFBS01 Fieldbus FTA Field Wiring

Model MU/MC-GFBS01 Fieldbus FTA Connection Diagram



Notes:

From GI Power Distribution Assembly

* HFIM = Honeywell Fieldbus Integrated MAU (Medium Attached Unit)
The GI/IS Power Supply Module is Honeywell part number 51192537-100 (MTL4053).
Connections for Segments 1 and 4 are shown. Segments 2 and 3 are identical.

17598-A

5. Fieldbus Service

5.1 MTL 4995 Fieldbus Power Supply Application Notes

Power supply function

The MTL4995 is designed to provide power for bus-powered devices on a fieldbus segment. The power supply is a plug-on module mounted on a Field Termination Assembly (FTA) on a per segment basis.

Output voltage

The output provided by the MTL4995 is regulated to approximately 19 volts and fully isolated from the incoming supply.

Terminations

An H1 fieldbus power supply, such as the MTL4995, which conforms to the later Clause 22 of the IEC 1158-2 fieldbus physical layer standard requires a defined output impedance characteristic over a range of frequencies covering both bit and message frame repetition rates. This is necessary because low-power signaling devices connected to the fieldbus can introduce change of direct current loading on the bus each time they change between receiving and a transmitting state. To avoid large oscillations on the bus when this occurs, the response of the overall bus system (comprising bus cables, terminators, and power supply) is defined to have no significant resonance peaks over the frequency range from 50 Hz to 40 kHz.

Signal balance requirement

The IEC 1158-2 standard also requires the fieldbus segment to be balanced about ground. This means that the fieldbus will normally be connected with neither cable grounded (unlike many of the traditional 4 – 20 mA analog systems installed today).

Current limit

The MTL4995 can supply a maximum of current to supply fieldbus devices. If an attempt is made to draw more current than 350 mA from the unit, either through an inadvertent short circuit during device connection or disconnection, or by adding too many devices to the bus, it will change to a constant current mode in which it delivers a fixed current just above its specified maximum output value. It will continue in this mode until the overload is removed, at which point it quickly returns to its normal operation. Repeated or continuous overloads will not damage the unit in any way.

5.2 MTL 4053 FF I.S. Isolator P. S. Application Notes

Isolator function

The MTL4053 is designed to provide power and communication to hazardous areas mounted devices on a fieldbus having a Physical Layer complying with IEC 1158-2 Clause 22. This means a fieldbus system having a 31.5 kbit/s data rate (commonly known as H1), with devices operating in either standard-power (defined by the earlier Clause 11 of IEC 1158-2) or low power signaling mode (defined by Clause 22). The most common system complying with these requirements is Foundation™ Fieldbus H1. The MTL4053 meets the combined requirements of a Type 33 Intrinsically Safe (I>S>) Power Supply and Galvanic Isolator in the Foundation Fieldbus 31.25 kbit/s Physical Layer Profile Specification (document reference FF-816).

FTA application

The MTL4053 is designed and certified as suitable for installation in a safe area or Division 2 hazardous area. In common with other MTL4000 series modules, the MTL4053 plugs into a Field Termination Assembly (FTA) which provides system and bulk power supply connections with a plug and socket format. Modules are retained in position on the FTA by two retaining screws.

Field wiring connections to the FTA are made through the screw terminals mounted on a blue, 6-terminal removable connector mounted on top of the module. The screw terminals accommodate stranded or solid wire up to 2.5 mm² (14 AWG).



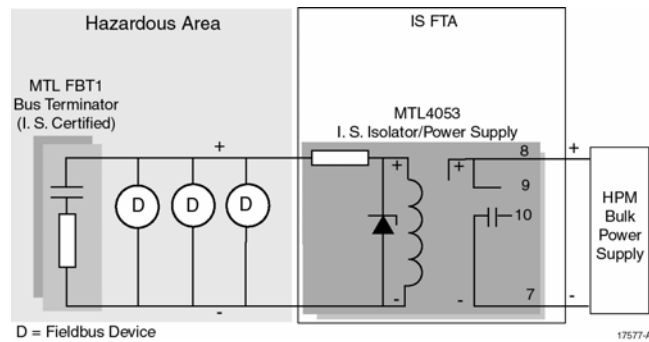
WARNING

When an FTA is installed or serviced in a Division 2 hazardous location, care should be taken to ensure that there is no explosion hazard present before connecting or disconnecting connection to non-i.s. circuits.

Operating characteristics

The IS FTA with the MTL4053 allows H1 segments to be extended into a hazardous area, to facilitate communication between these devices and others installed in the safe area of the segment.

Hazardous Area Fieldbus Using MTL4053 on the ISFTA



Save area devices

Safe area devices will normally be such items as operator display systems or bus configuration/monitoring stations that are powered separately from fieldbus. If bus powered devices are present on the safe area part of the fieldbus segment, they will require an additional fieldbus power supply (nonintrinsic safety FBFTA) to provide power for them. If this additional supply is not present (or if it does not include an internal bus terminator) then a separate fieldbus terminator unit may be required. Some general guidance to the installation of bus terminators with the I.S.FTA is provided in “Using the MTL4053 internal terminator” that follows, and the intrinsic safety aspects of the I.S.FTA together with any terminator installed in the hazardous area are covered in “Intrinsic Safety aspect of the I.S.FTA” that follows.

Isolation support

The MTL4053 provides full three-port isolation; that is, there is complete galvanic isolation between the hazardous area fieldbus connections, the safe area fieldbus connections and the incoming supply terminals. This isolation is important, as the IEC 1158-2 standard requires a fieldbus segment to operate in a balanced mode with respect to earth. Therefore, neither of the fieldbus cables will normally be connected to earth (unlike many of the 4-20 mA analog systems installed today).



WARNING

Only the cable shield are grounded and only grounded at the FTA connector screw.

It is therefore essential that an I.S. isolator/power supply provide connections that are fully isolated from each other and any other supplies. The MTL4053 provides this

Fieldbus Service

MTL 4053 FF I.S. Isolator P. S. Application Notes

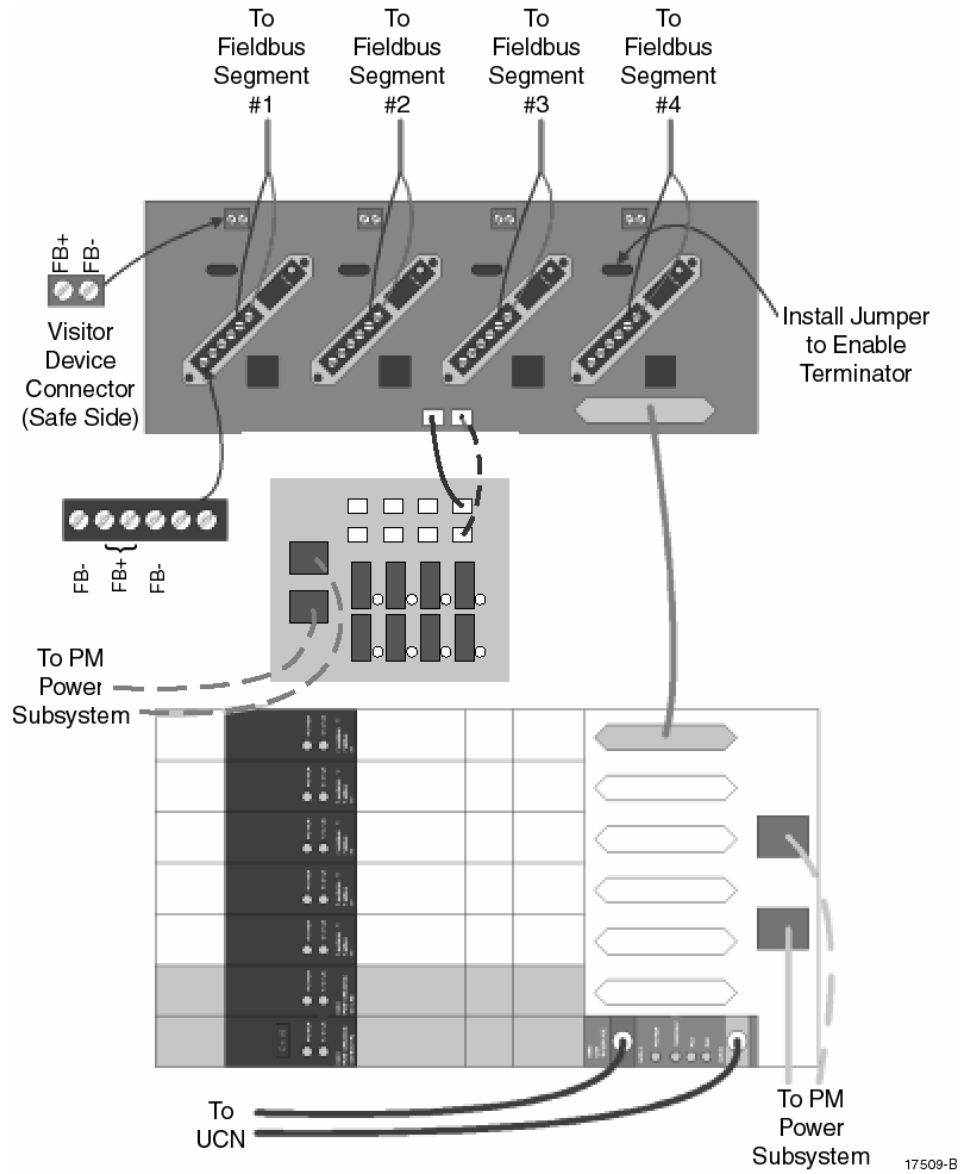
complete isolation (rated for 250 V and tested to 2400 V rms during manufacture), using specialized transformer coupling.

Installation areas

The MTL4053 will normally be installed in a safe area, but it is certified as suitable for installation in a Division 2 hazardous area (defined by North American standards). It may be connected at any convenient point on the fieldbus, bearing in mind the need to protect the I.S. circuits connected to hazardous area fieldbus terminals from any possibility of invasion by electric energy from other circuits. Care must be taken to choose the optimum connection point if the internal bus transmitter within the unit is connected. See the following figure.

Intrinsically safe installation

The I.S. output for connections to the devices located in a hazardous area fieldbus are shown in the following figure.



CAUTION

Since the MTL4053 is polarized and some devices to the fieldbus may sustain damage if the supply to them is reversed, care must be taken to ensure correct polarity of connections to the fieldbus.

Open circuit voltage

The power supply connected to these terminals provides an open circuit voltage of 18v. The power supply has a source resistance of approximately 105 Ω (formed by the current limiting resistor required for I.S. certification) and is electronically current limited at around 90 mA to protect the unit against intermittent and longer-term short circuit faults on the fieldbus.

Calculating current available at the end of the cable

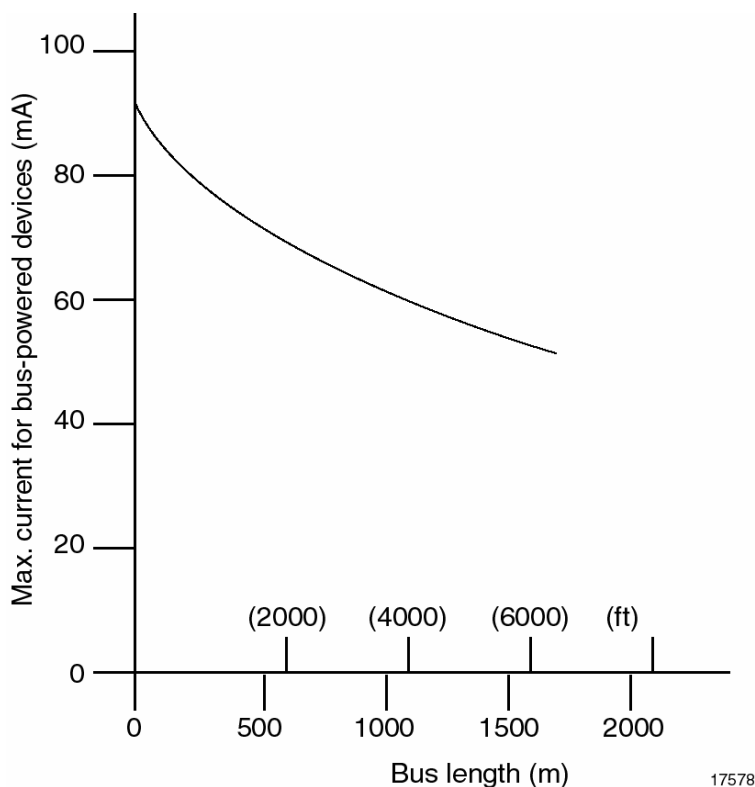
The current available for hazardous area devices depends on the length and resistance characteristics of the fieldbus cabling used. The IEC 1158-2 standard requires devices to operate with a terminal voltage down to 9 V. If devices are connected directly (with a very short bus cable length) to the hazardous area terminals on the ISFTA, the current available to them is defined simply by this minimum voltage together with the output voltage and the source resistance parameters of the MTL4053 circuit on the ISFTA. However, as the cable length to the hazardous area fieldbus devices is increased, its resistance becomes more important in determining the available current to devices at the far end of the bus. The calculation of this current (using Ohms Law) is obtained from:

$$\text{Available Current} = \frac{(\text{Open circuit output voltage of MTL4053}) - 9V}{(\text{Source resistance of MTL4053}) + (\text{Fieldbus cable loop resistance})}$$

Current characteristic illustration

The resulting typical characteristic when using Type A fieldbus cable (as defined for conformance testing in IEC 1158-2) is shown in the following figure.

Current Available for Hazardous Area Fieldbus Devices



Internal terminator

The ISFTA provides four sets of jumper pins that optionally activate an internal terminator inside each MTL4053 module. See the figure in Intrinsically safe installation. This terminator can be used for one of the two required terminations when the FTA is located at one end of the segment.

Fieldbus terminations

The IEC 1156-2 fieldbus Physical Layer standard (in Clauses 11 and 22) requires an H1 bus segment to have two fieldbus terminators installed. Careful consideration is needed to decide the optimum position to install the terminators. The standard specifies some simple rules that guarantee correct operation of the Physical Layer.

Fieldbus Service

MTL 4053 FF I.S. Isolator P. S. Application Notes

In summary, these are:

The maximum cable length between any two devices within one fieldbus segment should not exceed 1900 meters.

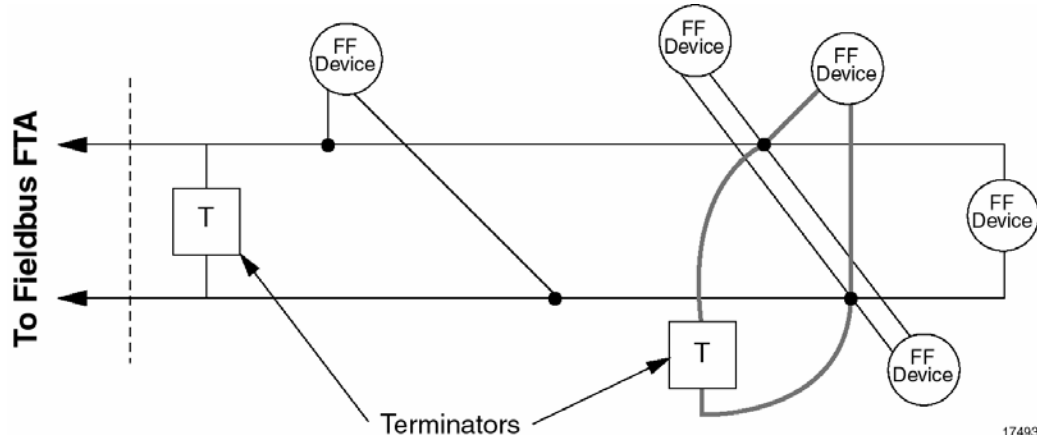
Two bus terminators (each consisting of a 100 ohm resistor and 1 μ F capacitor in series) should be placed on each end of the segment.

- The section of the fieldbus between the two bus terminators is called the main bus trunk cable.
- Cable runs to devices not connected with the main bus trunk are called spur cables. These cables have recommended lengths depending on the number of devices connected to them.

Placement determines length

The placement of terminators on the fieldbus trunk segment defines the position and length of the main bus trunk cable and any spur cables present.

General Fieldbus Segment Termination



Spur cables connecting to a single fieldbus device can generally be a maximum of 120 meters long (but shorter than this if they connect to multiple devices or the fieldbus segment includes more than 10 devices). If these limits, or the overall cable distance between devices, are exceeded an unacceptable level of communication errors may occur.

Normally, the inclusion of a significant lumped impedance (such as the current limiting resistor with the I.S. interface) would define that unit as one end of the main trunk. The

IEC 1158-2 standard specifically defines this in the case of an I.S. barrier unit, where a bus terminator must be fitted within 100 meters of the barrier.

Impedance balancing

However, in the case of a galvanically isolated I.S. interface such as the MTL4053, the use of active electronic circuits allows the value of the current limiting resistance (102 ohms in the MTL4053) to be compensated by using impedance balancing as far as fieldbus signaling is concerned. Thus the unit is almost transparent to fieldbus signals, allowing the bus on either side of it to be considered as a continuous part of the main bus trunk cable. Under dc and low frequency conditions, the current limiting resistor is, of course, present and defines the I.S. certification parameters of the unit and the current available for devices connected to the hazardous area portion of the bus.

I.S. certification

The MTL4053 unit is certified to European (CENELEC) and North American standards as suitable for connection to I.S. circuits in any degree of hazard (zone 0, 1, or 2) and gas mixture (gas group IIA IIB or IIC). The safety parameters for terminal connections comply with the FOUNDATION Fieldbus recommendations in document FF-816 and are:

Open circuit voltage:	22.0 V
Short circuit current:	214 mA (current limiting resistor 102 Ohms)
Matched output power:	1.19W

Any equipment connected to the safe area FTA must not be supplied from or contain voltages in excess of 250V rms or dc.

5.3 Basic Testing of Operating Segments

Types of network tests

Once the fieldbus network starts operation, several types of network tests can be performed.

- The simplest test is to determine if there is sufficient power on the wiring at each of the devices. This can be done with an ordinary digital voltmeter. Another simple test is to use the Receiver part of the Wire Tester to verify the general signal levels on the network.
- The next level of testing is to determine what devices are on the network and measure the signal amplitude of each device. Measuring the noise on the network is also useful. This can be done by with a Network Tester.

Fieldbus Service

Use of Communications Analyzer to Isolate Failures

- Beyond determining the health of the signals on the network, the information sent by the different devices to each other can be examined. This can be done by a computer that executes an analysis program. The computer acquires all frames on the network and shows their sequence on the screen. Discussion of protocol testing is beyond the scope of this Guide.

Generally, once a control network is operating properly, the fieldbus communications protocols do not break. Wiring, on the other hand, can deteriorate or be damaged in a number of ways. Having the capability to identify a wiring problem and to determine its location is very useful.

5.4 Use of Communications Analyzer to Isolate Failures

To be defined

5.5 Isolate Failures With Hand-Held Analyzer

Fieldbus Testers

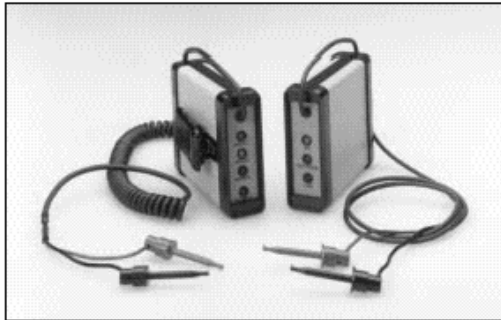
There are two general types of fieldbus network tests. One of these involves the content of the messages that are sent, what the messages mean, in what sequence the devices talk to each other, (that is, the protocols of the network). Once the protocol issues have been worked out, the network operation seldom needs to be examined. The other type of fieldbus test involves the wiring and signal transmission.

The testers described in the following discussion are intended to determine the ability of the wiring to carry fieldbus signals and to determine the health of the wiring and the devices transmitting signals on an operational fieldbus network. These are ongoing considerations because wiring can deteriorate.

5.6 Fieldbus Wire Tester, FBT-2

The fieldbus Wire Tester, FBT-2, is used to determine if newly installed or existing instrumentation wiring is usable for fieldbus. The FBT-2 is intended for installation and maintenance personnel and does not require detailed knowledge of fieldbus technology.

Fieldbus Wire Tester FBT-2



Wire tester operation

The Tester has two parts, a Transmitter and a Receiver. The Transmitter is typically used in the field end of the wiring. The Receiver is typically used in the control room end of the wiring.

Circuit to be Tested



The Transmitter and the Receiver have indicator lights that provide information about the polarity of the wiring and its condition.

When the Receiver and the Transmitter are attached to a wire pair correctly (the red probes to one wire; the black probes to the other wire), the Transmitters **OK** light is on. If the polarity is reversed, the Transmitters **REVERSE** light is on. This is useful for determining and documenting wiring polarity.

Before the Transmitter puts any signal on the wire pair, it first checks the cable to see if it is already carrying fieldbus signals. If the cable is carrying signals, the Transmitter will not produce a signal but will turn the **LIVE** indicator on. This prevents the Transmitter from interfering with an operating fieldbus network that it was connected to by mistake.

When the Receiver detects a fieldbus signal from the Transmitter that is above the level needed for fieldbus operation, the **GOOD** light is on. If the signal is at a higher level (the equivalent of less than 2 km of new fieldbus wire), the **EXCEL** indicator light is on (refer to Table 5-1). If the signal level is below that needed for fieldbus operation, the

SIGNAL light is on. This indicates that the Transmitter is connected to the wire pair but the signal level is too low. From these three indicator lights, it is possible to tell that the Receiver is attached to the right wire pair and if the signal on the wire pair being tested is high enough for fieldbus operation.

Test wire resistance with DVM

Additional wire tests need to be made using an ordinary digital voltmeter (DVM) to determine if a wire pair is suitable for fieldbus operation.

- Measure the resistance between each wire and the shield (if any) to verify that it is higher than 10K Ohms.
- Measure the resistance between the two twisted pair wires to verify that it is higher than 10 K Ohms.
- Measure the resistance between the shield (if any) and an earth ground to verify that the shield is not grounded.
- Short the wire pair on one end and measure the wire resistance. This information is needed when calculating the number of fieldbus devices that can be powered over the wires.

Both the Signal test and the Resistance tests assume that the wires are not connected to any fieldbus devices, terminators, power conditioners, etc.

Fieldbus Operation with FBT-2

The FBT-2 Receiver can also be used to check the general operation of a fieldbus network. When the Receiver is connected to an operating network, if the power on the network is greater than 10 V the POWER light comes on. The three signal lights indicate the signal strength on the network. If the signal is less than 175 mV pp, only the SIGNAL light is on. If the signal is greater than 175 mV pp, the level needed for fieldbus operation, the GOOD light is on. If the signal is well above the minimum needed, the EXCEL light is on.

Tester power

The Receiver operates from an internal battery or from the wall plug-in power supply. The power plug is also used to charge the battery. The Transmitter is powered by the Receiver over the wire pair being tested.

Tester specifications

- Model FBT-2A-A: fieldbus wire Tester with North American/Japanese style power plug.

- Model FBT-2E-A: fieldbus wire Tester with European style power plug.

Fieldbus Tester Specifications FBT-1,2

Min. signal detection level	50 mV pp +/- 10 mV
GOOD signal detection level	175 mV pp +/- 10 mV
EXCEL signal detection level	375 mV pp +/- 10 mV
Operation temp. range	-10 to 50° C
Weight 100 grams,	(0.25 lb.)each unit
Size	9 x 3 x 11 cm (3.5" x 1.2" x 4.3") each unit.

5.7 Fieldbus Network Tester, FBT-3

Network tester purpose

The fieldbus Network Tester, FBT-3, is used to examine a live fieldbus network without interfering with its operation. The Tester is intended for maintenance personnel to verify network operation or to troubleshoot an errant network.

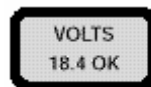
Fieldbus Network Tester, FBT-3



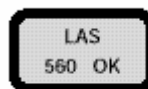
Examine Network parameters

A number of network parameters can be examined with the Tester. These are selected by pushing the SELECT button. This cycles the Tester through all its functions. When a function is selected, the measurement value shown initially is "???" until the needed data is gathered by the Tester. After that, the measured value is shown. The indication "OK" is shown if the measured value is within acceptable range.

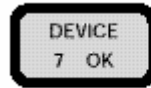
VOLTS: The DC voltage on the network is shown. Measurements over 9 volts are OK.



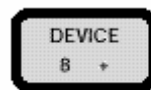
LAS: If there is any activity on the network, the Link Active Scheduler should be sending out Probe Node frames. The Tester measures the signal level of the Probe Node frame. The signal level is displayed in millivolts. Measurements over 150 mV are OK.



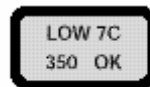
Device: If fieldbus devices are active on the network, the Tester counts them. If the count has remained the same since the Device function was selected, the display shows "OK."



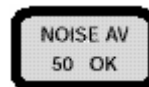
If a device has left the network the display shows a minus sign, "-", If a new one has been added, the display shows a plus sign, "+."



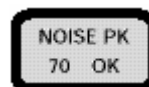
Low: The signal level of the device with the lowest signal level is shown. Measurements over 150 mV are OK. The device's address is shown behind the word LOW.



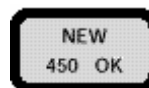
Noise Av: The noise on the network is measured in the silence period between frames. The value is averaged over 10 measurements. Measurements under 75 mV are OK.



Noise Pk: The peak noise recorded since the function was started is displayed. Measurements under 75 mV are OK.



New: If a new device is to be added to the network, it must respond to the Probe Node frame sent by the LAS. The Tester measures the signal level of the new device's response. Measurements over 150 mV are OK.



Fieldbus ServiceFieldbus Network Tester, FBT-3

Fieldbus tester Specifications FBT-3

The Tester is powered from the fieldbus itself and uses 5 mA of current. The Tester can be used in hazardous areas.

Operation temp. range	-10 to 50° C
Weight	100 grams, (0.25 lb.)
Size	9 x 3 x 11 cm (3.5" x 1.2" x 4.3")

6. Hazardous Area Classifications

6.1 Introduction

Reference Information

Information is provided to clarify the Hazardous Location installation requirements in North America and internationally. An explanation of the applicable enclosure classification systems is also provided.

6.2 North American Hazardous Location Standards

NEC and CEC Electrical Codes

Installation of electrical apparatus within hazardous (classified) locations of the United States is conducted under the provisions of the National Electrical Code (NEC), ANSI/NFPA 70, Article 500, and within Canada under the provisions of the Canadian Electrical Code (CEC) C22.1, Part 1, Section 18.

Classes

Hazardous (classified) locations, in both the United States and Canada, are categorized into three classes:

- **Class I** - Presence of flammable gases or vapors may be present in quantities sufficient to produce explosive or ignitable mixtures
- **Class II** - Presence of combustible dusts, powders or grains
- **Class III** - Presence of easily ignitable fibers or flyings

Divisions

The classes listed above are further categorized based upon the level of risk present:

- **Division 1** - Locations in which hazardous concentrations of flammable gases or vapors—or combustible dust in suspension—continuously, intermittently or periodically under normal operating conditions.
- **Division 2** - Locations in which flammable gases or vapors are present, but normally confined within closed containers or systems from which they can escape only under abnormal or fault conditions. Combustible dusts are not normally in suspension nor likely to be thrown into suspension.

Examples

Given the criteria above, the following examples are made:

- **Class III, Division 1** - A Class III, Division 1 location is a location in which easily ignitable fibers or material processing combustible flyings are handled, manufactured or used.
- **Class III, Division 2** - A Class III, Division 2 location is a location in which easily ignitable fibers are stored or handled.

Group Classifications

Flammable gases, vapors and ignitable dusts, fibers and flyings are classified into groups according to the energy required to ignite the most easily ignitable mixture within air. Group classifications are as follows:

Class I Group Classifications

Group A - Atmospheres containing acetylene

Group B - Atmospheres containing hydrogen, fuel and combustible process gases containing more than 30 percent hydrogen by volume, or gases or vapors of equivalent hazard.

Group C - Atmospheres such as ethyl ether, ethylene, or gasses or vapors of equivalent hazard.

Group D - Atmospheres such as acetone, ammonia, benzene, butane, cyclopropane, ethanol, gasoline, hexane, methanol, methane, natural gas, naphtha, propane or gases or vapors of equivalent hazard.

Class II Group Classifications

Group E - Atmospheres containing combustible metal dusts including aluminum, magnesium, and their commercial alloys, and other metals of similarly hazardous characteristics.

Group F - Atmospheres containing combustible carbonaceous dusts including carbon black, charcoal, coal or other dusts that have been sensitized by other materials so that they present an explosion hazard.

Group G - Atmospheres containing combustible dusts not included in Group E or F, including flour, wood, grain, and other dusts of similarly hazardous characteristics.

Designation	Permitted Use	Principle
XP	Division 1 & 2	Contains explosion and quenches flame
IS	Division 1 & 2	Limit energy of sparks under normal and fault conditions
Type X and Y	Division 1 & 2	Keeps flammable gas out
Type Z	Division 1 & 2	Keeps flammable gas out
NI	Division 1 & 2	No arcs, sparks or hot surfaces under normal conditions

Temperature Classification

Equipment intended for installation directly within the hazardous (classified) location must also be classified for the maximum surface temperature that can be generated under normal or fault conditions as referenced to either 40 °C or the maximum operating ambient of the equipment (whichever is greater). The maximum surface temperature must be less than the minimum autoignition temperature of the hazardous atmosphere present. The temperature shall be indicated in identification numbers as listed in Table A-1.

Temperature Identification Numbers (NEC/CEC)

Maximum Temperature		Identification Number
Degrees C	Degrees F	
450	842	T1
300	572	T2
280	536	T2A
260	500	T2B
230	446	T2C

Hazardous Area Classifications

North American Hazardous Location Standards

Maximum Temperature		Identification Number
Degrees C	Degrees F	
215	419	T2D
200	392	T3
180	356	T3A
165	329	T3B
160	320	T3C
135	275	T4
120	248	T4A
100	212	T5
85	185	T6

Intrinsically Safe Apparatus Parameters

V_{max} =	Maximum safe voltage which can be applied to the apparatus terminals.
I_{max} =	Maximum safe current which can be applied to the apparatus terminals.
C_i =	Unprotected capacitance in the apparatus which can be considered present at the terminals.
L_i =	Unprotected inductance in the apparatus which can be considered present at the terminals.
V_{oc} =	Maximum output voltage which can be delivered to the hazardous (classified) location. This voltage is the maximum from a single channel.
I_{sc} =	Maximum output current which can be delivered to the hazardous (classified) location. This current is the maximum from a single channel.

*Vt =	Maximum output voltage which can be delivered to the hazardous (classified) location. This voltage is the maximum across any combination of terminals of a multiple channel configuration.
*It =	Maximum output current which can be delivered to the hazardous (classified) location. This current is the maximum through any combination of terminals of a multiple channel configuration.
Ca =	Maximum capacitance which can be connected to the apparatus.
La =	Maximum inductance which can be connected to the apparatus.

*CSA does not recognize these parameters at this time

Entity Concept

Under entity requirements, the concept allows interconnection of intrinsically safe apparatus to associated apparatus, not specifically examined in such combination. The criterion for interconnection is that the voltage (V_{max}) and current (I_{max}), which intrinsically safe apparatus can receive and remain intrinsically safe, considering faults, must be equal to or greater than the voltage (V_{oc} or V_t) and current (I_{sc} or I_t) levels which can be delivered by the associated apparatus, considering faults and applicable factors.

In addition, the maximum unprotected capacitance (C_i) and inductance (L_i) of the intrinsically safe apparatus, including interconnecting wiring, must be less than or equal to the capacitance (C_a) and inductance (L_a) which can be safely connected to the associated apparatus. If these criteria are met, then the combination may be connected and remain intrinsically safe

6.3 International Electrotechnical Commission (IEC) Classifications

IEC Classification of Hazardous Locations

The IEC has established a number of recommendations applying to the construction of explosion protected electrical apparatus identified. These recommendations are found within IEC 79-0 through 79-15 and 79-28.

For all EC countries as well as various neighboring countries (CENELEC member states), the European Standards EN 50 014 to EN 50 020 and EN 50 039 apply for the construction of explosion protected electrical apparatus. They were established on the basis of the IEC Recommendations; however, in comparison they are much more detailed.

Hazardous Area Classifications

International Electrotechnical Commission (IEC) Classifications

Hazardous locations, as defined within IEC 7-10, are categorized into three zones:

- Zone 0 -** Explosive gas atmosphere is present continuously, or is present for long periods
- Zone 1 -** Explosive gas atmosphere is likely to occur in normal operation
- Zone 2 -** Explosive gas atmosphere is not likely to occur in normal operation and, if it does occur, it will exist for a short period only

Flammable gases, vapors and mists are classified into groups according to the energy required to ignite the most easily ignitable mixture within air. Apparatus is grouped according to the atmospheres it may be used within as follows:

- Group IIC -** Atmospheres containing acetylene, hydrogen, fuel and combustible process gases or vapors of equivalent hazard
- Group IIB -** Atmospheres such as ethyl ether, ethylene, or gasses or vapors of equivalent hazard
- Group IIA -** Atmospheres such as acetone, benzene, butane, cyclopropane, ethanol, gasoline, hexane, methanol, methane, natural gas, naphtha, propane or gases or vapors of equivalent hazard

Methods of Protection

Protection Concept	Designation	Permitted Use	Principle
Flameproof	d	Zone 1 & 2	Contains explosion and quenches flame
Intrinsic Safety	ia	Zone 0, 1 & 2	Limit energy of sparks under 2 faults
Intrinsic Safety	ib	Zone 1 & 2	Limit energy of sparks under 1 fault
Pressurized	p	Zone 1	Keeps flammable gas out
Encapsulation	m	Zone 1 & 2	Keeps flammable gas out
Increased Safety	e	Zone 1 & 2	No arcs, sparks or hot surfaces
Powder Filled	q	Zone 1 & 2	Contains explosion and quenches flame

Hazardous Area Classifications
International Electrotechnical Commission (IEC) Classifications

Oil Immersion	o	Zone 1 & 2	Keeps flammable gas out
non-sparking	nA	Zone 2	No arcs, sparks or hot surfaces under normal conditions
Enclosed Break	nC	Zone 2	Contains explosion and quenches flame
Limited Energy	nA	Zone 2	Limit energy of sparks and surface temperature under normal conditions
Restricted Breathing	nR	Zone 2	Keeps flammable gas out

Temperature Classification

Equipment intended for installation directly within the hazardous location must also be classified for the maximum surface temperature that can be generated under normal or fault conditions as referenced to the maximum operating ambient of the equipment. The maximum surface temperature must be less than the minimum autoignition temperature of the hazardous atmosphere present. The temperature shall be indicated in identification numbers as listed in the following table.

Temperature Identification Numbers (IEC)

Maximum Temperature		Identification Number
Degrees C	Degrees F	
450	842	T1
300	572	T2
200	392	T3
135	275	T4
100	212	T5
85	185	T6

6.4 Enclosure Ratings

NEMA and IEC Recognition

The NEMA (National Electrical Manufacturer's Association) enclosure classifications are recognized in the US. The IEC Publication 529 Classifications are recognized throughout Europe and those parts of the world that use the IEC standards as a basis for product certifications. The following paragraphs provide a discussion of the Comparison Between NEMA Enclosure Type Numbers and IEC Enclosure Classification Designations.

IEC Classifications

IEC Publication 529, Classification of Degrees of Protection Provided by Enclosures, provides a system for specifying the enclosures of electrical equipment on the basis of the degree of protection provided by the enclosure. IEC 529 does not specify degrees of protection against mechanical damage of equipment, risk of explosion, or conditions such as moisture (produced for example by condensation), corrosive vapors, fungus, or vermin.

NEMA Standards

NEMA Standards Publication 250, Enclosures for Electrical Equipment (1000 Volts Maximum), does test for environmental conditions such as corrosion, rust, icing, oil, and coolants. For this reason, and because the tests and evaluations for other characteristics are not identical, the IEC enclosure classification designations cannot be exactly equated with NEMA enclosure type numbers.

IEC Designations

Basically, the IEC designation consists of the letters IP followed by two numerals. The first characteristic numeral indicates the degree of protection provided by the enclosure with respect to persons and solid foreign objects entering the enclosure. The second characteristic numeral indicates the degree of protection provided by the enclosure with respect to the harmful ingress of water.

The following table provides an approximate conversion from NEMA enclosure type numbers to IEC enclosure classification designations. The NEMA types meet or exceed the test requirements for the associated IEC classifications; **for this reason, the Table cannot be used to convert from IEC classifications to NEMA types.**

NEMA Enclosure Type Number	IEC Enclosure Classification Designation
1	IP 10
2	IP 11
3	IP 54
3R	IP 14
3S	IP 54
4 and 4X	IP 56
5	IP 52
6 and 6P	IP 67
12 and 12K	IP 52
13	IP 54

NOTE: *This comparison is based on tests specified in IEC Publication 529*

6.5 Table III Options Reference

Codes and Descriptions

The following table lists available ST 3000 FF Table III approval body options alphabetically and numerically by their codes and gives a brief description of the options. Note that restrictions do apply based on other as-built transmitter characteristics and some options are mutually exclusive.

ST 3000 FF Table III Approval Body Options

If Code is . . .	Then transmitter option is . . .
1C	FM approval body certification for: Explosion proof/Flameproof Class I, Division 1, Groups A, B, C, D Dust Ignition Proof Class II, III, Division 1, Groups E, F, G Non-Incendive Class I, II, III Division 2, Groups A, B, C, D, E, F, G Intrinsically Safe Class I, II, III, Division 1, Groups A, B, C, D, E, F, G
2J	CSA approval body certification for: Explosion proof Class I, Division 1, Groups B, C, D Dust Ignition Proof Class II, III, Division 1, Groups E, F, G Intrinsically Safe Class I, II, III, Division 1, Groups A, B, C, D, E, F, G Suitable for use in Class I, II, III, Division 2, Groups A, B, C, D, E, F, G
3A	LCIE approval body certification for: Flame Proof/ CENELEC EEx d IIC T6 Intrinsically Safe/CENELEC EEx ia IIC T5
3N	Zone 2 (Europe) certification for: Self-Declared per 94/9/EC (ATEX4) Ex II 3 GD T6 X
4H	SA approval body certification for: Intrinsically Safe Ex ia IIC T4 Non-Incendive Ex n IIC T6 (T4 with Local Smart Meter option) Flameproof Ex d IIC T6

Reader Comments

Fax Transmittal		Fax No.: (602) 313-4212
Reader Comments		
To:	Information Development	
From:	Name	Date:
	Title:	
	Company:	
	Address:	
	City:	State: Zip:
	Telephone:	Fax:
Fieldbus Planning, Installation, and Service Manual, FB02-610		
Release R610, 09/00		
Comments:		
You may also call 800-343-0228 (available in the 48 contiguous states) or write to:		
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